

THE BULLETIN

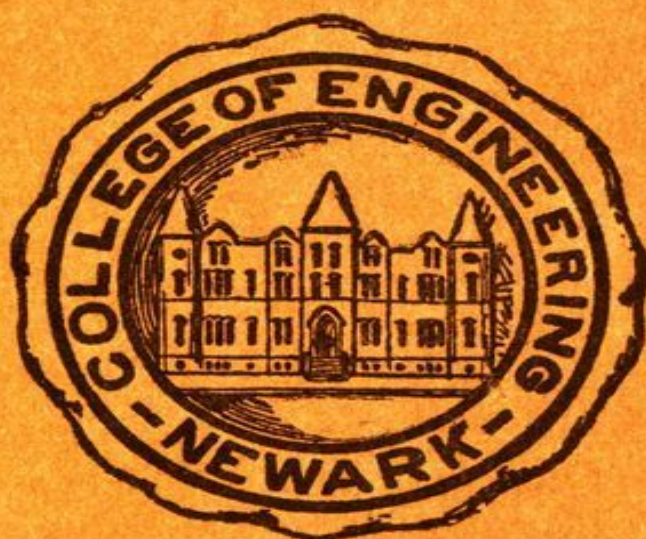
VOLUME XI

DECEMBER 15, 1938

NUMBER 4

The Newark College of Engineering

CATALOGUE NUMBER



365 to 369 High Street

Newark, New Jersey

THE BULLETIN

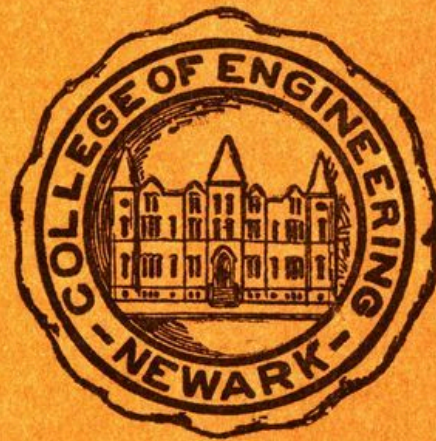
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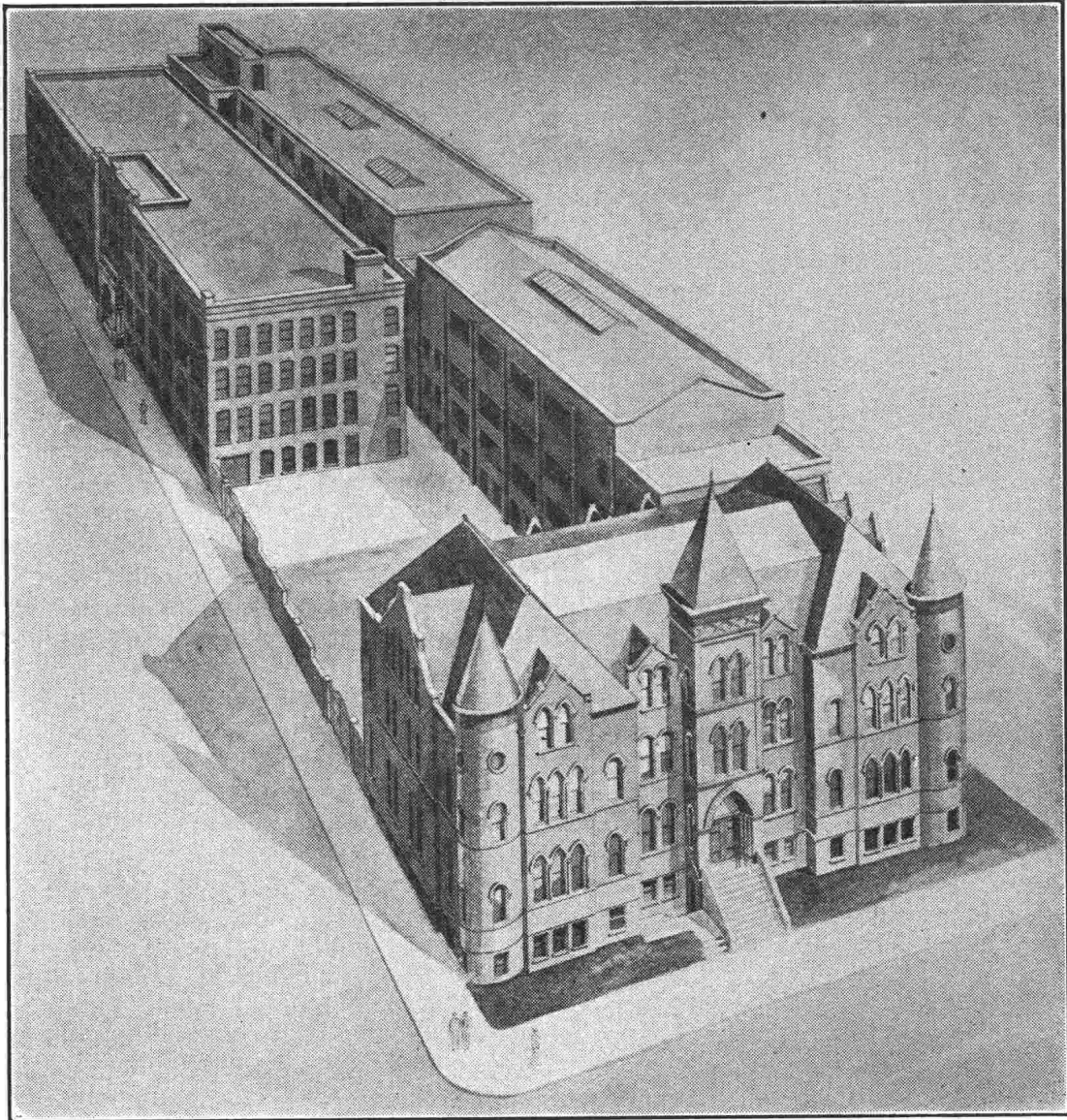
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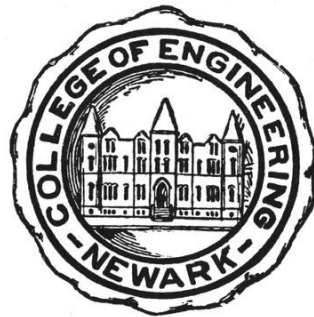
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COLLEGE OF ENGINEERING BULLETIN, V. 11, NO. 4, DECEMBER
15, 1938 ***



THE NEWARK COLLEGE OF ENGINEERING

The Newark College of Engineering NEWARK, NEW JERSEY



1939-1940

365-369 HIGH STREET
NEWARK, NEW JERSEY

COLLEGE CALENDARS^[1]

1939

First Semester Classes End (Freshmen)	January	20
Mid-year Examinations (Freshmen)	Jan. 23 -	Feb. 4
First Semester Classes End (Sophomores, Juniors and Seniors)	January	27
Mid-year Examinations (Sophomores, Juniors and Seniors)	Jan. 30 -	Feb. 4
Second Semester Begins	February	6
Visitors' Day	February	11
Lincoln's Birthday	February	13
Washington's Birthday	February	22
Good Friday	April	7
Spring Recess	April	10-15
Re-examinations	April	10-15
Memorial Day	May	30
Second Semester Classes End (Freshmen, Sophomores and Juniors)	June	2
Final Examinations (Freshmen, Sophomores and Juniors)	June	5-10
Second Semester Classes End (Seniors—No Final Examinations)	June	9
Re-examinations	September	5-9

Entrance Examinations	September	5-9
Registration (Freshmen)	{ September	11 to
	{ Sept.	13 at noon
Registration (Sophomores, Juniors and Seniors)	September	11-16
College Opens	September	18
Thanksgiving Recess	{ Nov. 29 at noon	
	{ to Dec. 4 at 9 A. M.	
Christmas Recess	{ Dec. 22 at noon	
	{ to Jan. 2 at 9 A. M.	

1940

First Semester Classes End (Freshmen)	January	19
Mid-year Examinations (Freshmen)	Jan. 22 -	Feb. 3
First Semester Classes End (Sophomores, Juniors and Seniors)	January	26
Mid-year Examinations (Sophomores, Juniors and Seniors)	Jan. 29 -	Feb. 3
Second Semester Begins	February	5
Visitors' Day	February	10
Lincoln's Birthday	February	12
Washington's Birthday	February	22
Good Friday	March	22
Spring Recess	April	8-13
Re-examinations	April	8-13
Memorial Day	May	30

Second Semester Classes End (Freshmen, Sophomores and Juniors)	May	31
Final Examinations (Freshmen, Sophomores and Juniors)	June	3-8
Second Semester Classes End (Seniors—No Final Examinations)	June	7
Re-examinations	September	3-7
Entrance Examinations	September	3-7
Registration (Freshmen)	{ September 9 to { Sept. 11 at noon	
Registration (Sophomores, Juniors and Seniors)	September	9-14
College Opens	September	16
Thanksgiving Recess	{ Nov. 27 at noon to { Dec. 1 at 9 A. M.	
Christmas Recess	{ December 23 to { Jan. 2 at 9 A. M.	

[1] For calendars for individuals beginning the work of the Freshman year in February 1939 or February 1940 see [page 7](#).

ANNOUNCEMENT

OPPORTUNITIES FOR STUDENTS TO BEGIN THE WORK OF THE FRESHMAN YEAR IN FEBRUARY

In order to meet a very real need the College has for the past several years made it possible for a limited number of high school graduates to begin the work of the Freshman year in February. Men entering at that time may, if they so desire, continue the work of the Freshman year during the Summer Session. Those who satisfactorily complete the work of the Spring and Summer Sessions are then eligible for admission to the Sophomore class the following September.

A man who enters in February, therefore, is able to graduate one year earlier than would have been possible if he had waited until the following September to enter. Freshmen continuing throughout the Spring and Summer Sessions receive the same number of credit hours of instruction as do those students who take the regular Freshman program beginning in September. The number of class hours of attendance per week for the February group is somewhat greater than the weekly total for those entering in September. However, experience has shown that the better student is able to complete the work of the Spring and Summer Sessions satisfactorily without suffering any ill effects. As the Summer Session ends about the middle of August, a vacation of approximately one month is available between the closing of the Summer Session and the beginning of classes in September.

High school students who wish to apply for admission to the February sections should submit certificates of high school training immediately after the December 1st preceding the February entrance as the enrollment is definitely limited. These transcript and application forms may be obtained upon request from the College Registrar. While it is understood that a complete secondary school record will usually not be available until the student's graduation, the partial record will serve for preliminary consideration of the application. The final statement will be obtained by the

College. All certificates of high school training must be mailed directly to the Registrar by the principals of the high schools.

Expenses for the February group are the same as for the September matriculates.

Individuals desiring additional information may apply to the Registrar for an appointment.

COLLEGE CALENDARS

FEBRUARY ADMISSIONS to the FRESHMAN CLASS

1939

Entrance Examinations	Jan. 30 -	Feb. 1
Registration for Spring Session	Jan. 30 -	Feb. 4
Spring Session Begins	February	6
Visitors' Day	February	11
Lincoln's Birthday	February	13
Washington's Birthday	February	22
Good Friday	April	7
Spring Recess	April	10-15
Memorial Day	May	30
Spring Session Ends	June	2
Examinations	June	5-10
Registration for Summer Session	June	5-10
Summer Session Begins	June	12
Independence Day	July	4
Summer Session Ends	August	18
Re-examinations	September	5-9

1940

Entrance Examinations	January	29-31
Registration for Spring Session	{ January	29 to
	{ January	31 at noon
Spring Session Begins	February	5
Visitors' Day	February	10
Lincoln's Birthday	February	12
Washington's Birthday	February	22
Good Friday	March	22
Spring Recess	April	8-13
Memorial Day	May	30
Spring Session Ends	May	31
Examinations	June	3-8
Registration for Summer Session	June	3-8
Summer Session Begins	June	10
Independence Day	July	4
Summer Session Ends	August	16
Re-examinations	September	3-7

1939

JANUARY							JULY						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
1	2	3	4	5	6	7	--	--	--	--	--	--	1
8	9	10	11	12	13	14	2	3	4	5	6	7	8
15	16	17	18	19	20	21	9	10	11	12	13	14	15
22	23	24	25	26	27	28	16	17	18	19	20	21	22
29	30	31	--	--	--	--	23	24	25	26	27	28	29
--	--	--	--	--	--	--	30	31	--	--	--	--	--
FEBRUARY							AUGUST						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	--	--	1	2	3	4	--	--	1	2	3	4	5
5	6	7	8	9	10	11	6	7	8	9	10	11	12
12	13	14	15	16	17	18	13	14	15	16	17	18	19
19	20	21	22	23	24	25	20	21	22	23	24	25	26
26	27	28	--	--	--	--	27	28	29	30	31	--	--
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MARCH							SEPTEMBER						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	--	--	1	2	3	4	--	--	--	--	--	1	2
5	6	7	8	9	10	11	3	4	5	6	7	8	9
12	13	14	15	16	17	18	10	11	12	13	14	15	16
19	20	21	22	23	24	25	17	18	19	20	21	22	23
26	27	28	29	30	31	--	24	25	26	27	28	29	30
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APRIL							OCTOBER						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	--	--	--	--	--	1	1	2	3	4	5	6	7
2	3	4	5	6	7	8	8	9	10	11	12	13	14
9	10	11	12	13	14	15	15	16	17	18	19	20	21
16	17	18	19	20	21	22	22	23	24	25	26	27	28
23	24	25	26	27	28	29	29	30	31	--	--	--	--
30	--	--	--	--	--	--	--	--	--	--	--	--	--
MAY							NOVEMBER						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	1	2	3	4	5	6	--	--	--	1	2	3	4
7	8	9	10	11	12	13	5	6	7	8	9	10	11
14	15	16	17	18	19	20	12	13	14	15	16	17	18
21	22	23	24	25	26	27	19	20	21	22	23	24	25
28	29	30	31	--	--	--	26	27	28	29	30	--	--
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JUNE							DECEMBER						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
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4	5	6	7	8	9	10	3	4	5	6	7	8	9
11	12	13	14	15	16	17	10	11	12	13	14	15	16
18	19	20	21	22	23	24	17	18	19	20	21	22	23
25	26	27	28	29	30	--	24	25	26	27	28	29	30
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1940

JANUARY							JULY						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	1	2	3	4	5	6	--	1	2	3	4	5	6
7	8	9	10	11	12	13	7	8	9	10	11	12	13
14	15	16	17	18	19	20	14	15	16	17	18	19	20
21	22	23	24	25	26	27	21	22	23	24	25	26	27
28	29	30	31	--	--	--	28	29	30	31	--	--	--
--	--	--	--	--	--	--	--	--	--	--	--	--	--
FEBRUARY							AUGUST						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	--	--	--	1	2	3	--	--	--	--	1	2	3
4	5	6	7	8	9	10	4	5	6	7	8	9	10
11	12	13	14	15	16	17	11	12	13	14	15	16	17
18	19	20	21	22	23	24	18	19	20	21	22	23	24
25	26	27	28	29	--	--	25	26	27	28	29	30	31
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MARCH							SEPTEMBER						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	--	--	--	--	--	1	1	2	3	4	5	6	7
3	4	5	6	7	8	9	8	9	10	11	12	13	14
10	11	12	13	14	15	16	15	16	17	18	19	20	21
17	18	19	20	21	22	23	22	23	24	25	26	27	28
24	25	26	27	28	29	30	29	30	--	--	--	--	--
31	--	--	--	--	--	--	--	--	--	--	--	--	--
APRIL							OCTOBER						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	1	2	3	4	5	6	--	--	1	2	3	4	5
7	8	9	10	11	12	13	6	7	8	9	10	11	12
14	15	16	17	18	19	20	13	14	15	16	17	18	19
21	22	23	24	25	26	27	20	21	22	23	24	25	26
28	29	30	--	--	--	--	27	28	29	30	31	--	--
--	--	--	--	--	--	--	--	--	--	--	--	--	--
MAY							NOVEMBER						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	--	--	1	2	3	4	--	--	--	--	--	1	2
5	6	7	8	9	10	11	3	4	5	6	7	8	9
12	13	14	15	16	17	18	10	11	12	13	14	15	16
19	20	21	22	23	24	25	17	18	19	20	21	22	23
26	27	28	29	30	31	--	24	25	26	27	28	29	30
--	--	--	--	--	--	--	--	--	--	--	--	--	--
JUNE							DECEMBER						
S	M	T	W	T	F	S	S	M	T	W	T	F	S
--	--	--	--	--	--	1	--	--	--	--	--	1	2
2	3	4	5	6	7	8	3	4	5	6	7	8	9
9	10	11	12	13	14	15	10	11	12	13	14	15	16
16	17	18	19	20	21	22	17	18	19	20	21	22	23
23	24	25	26	27	28	29	24	25	26	27	28	29	30
30	--	--	--	--	--	--	31	--	--	--	--	--	--

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ALLAN R. CULLIMORE
President

JAMES A. BRADLEY
Dean

HAROLD N. CUMMINGS
Supervisor of Evening Sessions

L. C. SPRY
[\[2\]](#) *Secretary to the Faculty*

H. H. METZENHEIM *Comptroller*

R. W. VAN HOUTEN *Assistant to the President*

LILLIAN M. SCOTT *Bursar*

P. L. CAMBRELENG *Registrar*

MARGARET A. YATSKO *Recorder*

C. H. STEPHANS *Supt. of Buildings and Equipment*

E. B. BERLINRUT *Director of Publicity*

GERTRUDE C. ISAACS *Secretary to the President*

EDNA SCHNEIDER *Asst. to Sec'y of Faculty*

ADELE GARRISON *Asst. to Bursar*

[\[2\]](#) All communications to the Faculty should be addressed to the Secretary.

COLLEGE LIBRARY

FACULTY COMMITTEE

P. M. GIESY, Chairman

J. JOFFE

J. M. ROBBINS

G. D. WILKINSON

STAFF

KATHARINE MAYNARD

Consulting Librarian

P. M. GIESY

Librarian

RUTH LITTIG

Librarian in Charge of Circulation

GLADYS E. BIRKELO

Cataloguer

MARION PAGE

Assistant Librarian

C. P. DEUTSCH

Assistant Librarian

The Library of the College consists of approximately 21,000 volumes of technical reference books and of engineering texts, together with many volumes covering all branches of literature, and bound volumes of the more important engineering periodicals. The College subscribes to a considerable number of engineering periodicals and trade journals which are available on the shelves of the library for student use. In addition the College has an arrangement with the Newark Public Library so that books may upon request be obtained from the city library for reference and general use. The facilities of the Library of the Public Service Corporation are also at the service of the College.

In 1936 Dr. Edward Weston bequeathed to the College his scientific library of approximately 12,000 bound volumes and 65,000 pamphlets, together with his collection of scientific and laboratory apparatus. Arrangements are

now in progress to provide adequate facilities for making these collections available to the students and the public.

THE NEWARK COLLEGE OF ENGINEERING

The Newark College of Engineering instituted in 1919, is a development of the Newark Technical School founded in 1881 by the Board of Trade of Newark. The College is a public institution supported by both the City and the State and is governed by a Board of Trustees appointed by the Governor of the State of New Jersey. The Governor and the Mayor of Newark are ex-officio members.

The control and supervision of finances of the College is vested in the Board of Regents of the State of New Jersey.

The College is situated in the heart of Newark at High Street and Summit Place. The work of the institution is carried on in four buildings. The recitation halls are adequately equipped modern structures, particularly suited to requirements of an engineering college.

Located in the center of one of the most important industrial sections in the world, the opportunities for direct contact with industry are exceptionally good and co-operative relations have been established with some of the largest and best industries in this section.

GENERAL INFORMATION

Industry is asking today for young men of character and initiative who have been trained to leadership along the lines of commercial production. Men are in great demand who can step into positions of more or less influence and who can handle problems of manufacture as well as problems of design. The work of the production engineer of today is concerned with problems of labor and problems of money as well as with problems having to do with materials.

The Administration of the College believes that success in the field of engineering depends, today, upon certain fundamental factors which are best taught in direct contact with modern industry. The factors are:

- (1) Character
- (2) Initiative
- (3) Hard work
- (4) Understanding human relations
- (5) A knowledge of fundamentals of applied science.

Early contact with the industries enables the student to get not only his academic work but also to learn at first-hand some very important and fundamental things about the operation of modern industry and about the functions of the modern engineer.

Beginning in the year 1940-41 two full semesters of academic work will be given to senior students. The co-operative work for students affected by this arrangement will be increased and it will be given during the summers following the Sophomore and Junior years. This work will take the place of the alternating schedule now in effect for seniors.

The summer cooperative work will have the same purpose as that heretofore given in the Senior year. It will serve as an industrial engineering laboratory where the men will work under commercial conditions, commercial standards and commercial criteria. The work given in industry under the supervision of the College will, however, be limited to those men who have shown maturity, accomplishment, and development in the first two years of their College work. It will be in the nature of a premium given

those men who are likely to profit from it, along with other premiums in the way of scholarships, exemptions from examinations, etc. extended to an Honors Option group. It will be optional on the part of the student and limited by the College.

The college is particularly interested in the study of traits and characteristics which in individual cases interfere with the student's development along industrial and professional lines. The student is given, therefore, a considerable amount of individual study and attention from this particular point of view.

COURSES

The College offers four-year courses in Industrial Chemistry, and in Civil, Electrical and Mechanical Engineering, with an option in Aeronautical Engineering.

Much of the subject matter in these four courses is common to all of them. These common subjects represent the unity of the basic sciences and techniques of all branches of engineering.

The hours of instruction extend from 9:00 A. M. to 5:00 P. M. Monday to Friday inclusive.

Opportunity is offered to a limited number of high school graduates to enter the College in February as freshmen. Students entering at that time may continue the work of the freshman year throughout the summer. Those who satisfactorily complete the work of the spring and summer terms may be admitted to the sophomore class in September of the same year.

Students other than those referred to in the above paragraph, who are desirous of taking work during the summer, should inquire at the College for information regarding courses to be given.

DEGREES

The Newark College of Engineering reports to and receives the right to grant degrees from the New Jersey State Board of Education.

All candidates for graduation who satisfactorily complete a regular course of study and the examinations required receive the degree of Bachelor of Science (B.S.) in the course pursued. The degree is certified by a diploma bearing the seal and signatures of officers of this institution.

The work of the Newark College of Engineering is accredited by the Middle States Association of Colleges and Secondary Schools, and by the American Council on Education.

The courses of this college are registered by The State Education Department of the University of the State of New York.

National professional engineering societies sponsor student branches at the College. The societies represented are:

American Institute of Chemical Engineers

American Society of Civil Engineers

American Institute of Electrical Engineers

American Society of Mechanical Engineers

Society for the Advancement of Management.

The specific academic requirements in the four courses are shown on the next following pages for students entering September 1937 and later.

Graduates of 1940 meet similar requirements as stated in earlier issues of this bulletin.

One Credit Hour is, generally, equivalent to one hour of attendance per week in class or lecture exercises, and is equivalent to two hours of attendance per week in laboratory exercises, during a semester.

COMMON AND NON-PROFESSIONAL SUBJECTS

Required in All Courses

	SUBJECT	Credit Hours
CH	11 Chemistry	9
CH	21 Qualitative Analysis	5
EE ^[3]	21 Electricity	6
ME	1 Engineering Drawing	5
ME	2 Engineering Drawing	2
Eng	10 English	8
Eng	20 English	6
Eng	50 History of Industrial Civilization	2
Eng	60 History of Industrial Civilization	2
Ind E ^[4]	11 Principles of Engineering (The College)	4
Ind E	12 Principles of Engineering (The Industry)	2
Ind E	31 Economics	3
Ind E	41 Accounting	3
Math	1 Mathematics	7.5
Math	21 Calculus	8
Mech ^[5]	20 Mechanics	4
Mech ^[5]	21 Mechanics	4
Phys	1 Introductory Problems in Physics	2.5

Phys	2 Physics	3.5
Phys	3 Physics	7.5

[3] EE 71 or EE 75 may be substituted for EE 21.

[4] Required of students entering September, 1938, and later.

[5] Mech 22 or Mech 23 may be substituted for Mech 21.

Mech 24 may be substituted for both Mech 20 and Mech 21.

PROFESSIONAL AND TECHNICAL SUBJECTS

Civil Engineering Course

SUBJECT			Credit Hours
CE	1	Surveying	8
CE	2	Surveying	10
CE	10	Sanitation	6
CE	11	Sanitation	6.5
CE	20	Highways	3
CE	21	Highways	2
CE	22	Highway Traffic Control	2
CE	30	Structures	10
CE	40	Hydraulics	4.5
ME	31	Thermodynamics	3
ME	55	Mechanical Engineering	4.5
Ind E	13	Staff Control	3
Ind E	14	Staff Control	4
Ind E	22	Industrial Management	3
Ind E	51	Business Law	1
Phys	30	Strength of Materials	7.5

Electrical Engineering Course

SUBJECT		Credit Hours
CE	41 Hydraulics	3
EE	22 Electric Circuits	7.5
EE	31 Electric Networks	2
EE	32 Electric Transients	2
EE	33 Electric Machinery	8
EE	35 Electron Tubes	5
EE	41 Electric Transmission Equipment	2
EE	42 Electric Transmission Circuits	3
EE	43 Electric Machinery	7
EE	45 Electrical Measurements	3.5
EE	46 Electron Tube Circuits	2.5
EE	47 Electrical Design	3.5
ME	16 Machine Design	3
ME	31 Thermodynamics	3
ME	55 Mechanical Engineering	4.5
Ind E	13 Staff Control	3
Ind E	14 Staff Control	4
Ind E	22 Industrial Management	3
Ind E	51 Business Law	1
Phys	30 Strength of Materials	7.5

Industrial Chemistry Course

SUBJECT		Credit Hours
CH	22 Inorganic Chemistry	4

CH	31	Physical Chemistry	5
CH	32	Quantitative Analysis	11
CH	33	Thermodynamics	4
CH	41	Physical Chemistry	6
CH	42	Organic Chemistry	9
CH	43	Industrial Chemistry	3.5
EE	83	Applied Electricity	4.5
ME	16	Machine Design	3
ME	55	Mechanical Engineering	4.5
Ind E	13	Staff Control	3
Ind E	14	Staff Control	4
Ind E	22	Industrial Management	3
Ind E	51	Business Law	1
Phys	30	Strength of Materials	7.5

Mechanical Engineering Course

SUBJECT			Credit Hours
CE	41	Hydraulics	3
EE	81	Applied Electricity	9
ME	7	Shop Practice	1.5
ME	10	Mechanisms	6
ME	14	Machine Design	9
ME ^[6]	18	Graphics & Structural Design	4.5
ME	20	Physical Metallurgy	3
ME	22	Metallography	1.5
ME	30	Thermodynamics	4

ME	34 Heat Power	4.5
ME ^[6]	36 Power Plants	3
ME ^[6]	37 Applied Heat Power	3
ME ^[6]	50 Mechanical Laboratory	4
Ind E	13 Staff Control	3
Ind E	14 Staff Control	4
Ind E	21 Industrial Management	5
Ind E	51 Business Law	1
Phys	30 Strength of Materials	7.5

^[6] In the Aeronautical Option the following subjects are required in place of those marked (6):

ME	90 General Aeronautics	6
ME	91 Airplane Structure	6
ME	92 Airplane Engines	2.5

While passing marks are required as a minimum in all subjects undertaken, barely passing marks alone do not insure graduation.

To be eligible for graduation, a student is required to attain a grade of A or B in at least 20% of the credit hours allotted in the catalogue to professional and technical subjects.

All graduates of the College who desire to become candidates for the degree C. E., E. E., or M. E., must receive the approval of the faculty at least eighteen months before the granting of the degree.

Each candidate for the above-mentioned degrees shall render every three months to the head of the department of which he is a graduate, a written report on his progress, such report to contain a brief outline of the engineering work performed by the candidate, the names of engineering books and articles read by him, and the list of engineering society meetings which he has attended.

Each candidate shall submit to the faculty, at least four months before the granting of the degree, a satisfactory thesis upon an approved subject.

Each candidate shall appear in person upon the appointed commencement day to receive his degree, unless excused by the faculty.

ADMISSION TO THE COLLEGE

A student may enter the College of Engineering as a matriculated student, registered as a candidate for a degree, or as a special student, permitted to attend such courses in the College as he may be qualified to take, but not as a candidate for a degree.

Applicants may submit certificates of secondary school records to the College Registrar after March 15 for the next September opening of College or after December 1 for the next February opening. The forms for certification will be provided on request. While it is understood that a complete secondary school record will not usually be available until later, the partial record will serve for preliminary consideration of the application.

It is required that each applicant present himself for an interview at a time arranged by the Registrar so that the College may evaluate his probable fitness to do engineering college work and subsequently to find employment in the profession. This estimate will be on the basis of physical and emotional fitness and on previous scholastic achievement. In instances where the evidence is not reasonably conclusive, certain tests and examinations may be required. A fee will be charged for this testing service.

Every applicant for entrance into the Freshman Class must furnish to the College a statement of good moral character.

Class room and laboratory facilities, demands of good instruction, and prospects of employment in a particular field limit the number of students to be admitted each year. It was found necessary, in view of these factors, to close the September 1938-39 admissions in Industrial Chemistry on August 15, 1938, and to require that the matriculation fee be paid at that time. There is a possibility that this and other courses may be affected by similar closing dates and that the matriculation fee may be payable on notification by the Registrar prior to the registration dates published in this catalog.

It is requested that all Freshmen complete their registration arrangements for September 1939-40 admissions before Wednesday, September 13, 1939, at twelve noon. An extra registration fee will be required of those who register after that time. Similarly, February admissions will be closed at noon on the Wednesday of registration week, and the extra registration fee will be held effective after that time.

ENTRANCE SUBJECTS

All candidates for matriculation must offer eight entrance units in the following required subjects:

SUBJECT	College Entrance Examination Board Equivalents
English—4 years	3 units English
Elementary Algebra	1 unit Mathematics A1
Intermediate Algebra	½ unit Mathematics A2
Plane Trigonometry	½ unit Mathematics E or Gamma
Plane Geometry	1 unit Mathematics C
Physics	1 unit Elementary Physics
Chemistry, or	} 1 unit Elementary Chemistry Elementary Biology —
Biology, or	
General Science	

and at least seven entrance units in the following elective subjects:

Academic Subjects

Latin	1, 2, 3 or 4 units Latin 2, 3A, 3B, 4, H, K
German	1, 2, 3 or 4 units German 2, 3 or 4
French	1, 2, 3 or 4 units French 2, 3 or 4
Spanish	1, 2, 3 or 4 units Spanish 2, 3 or 4
Italian	1, 2, 3 or 4 units ———

History	1, 2, 3 or 4 units	History A, B, C or D
Adv. Algebra	$\frac{1}{2}$ unit	Mathematics B
Solid Geometry	$\frac{1}{2}$ unit	Mathematics D
Economics	1 unit	——

While languages are listed here as electives, applicants are advised that French or German or both are usually required for graduate work.

Commercial and Vocational Subjects^[7]

NO C.E.E.B. EQUIVALENTS

Drawing	1 unit
Electricity	$\frac{1}{2}$ or 1 unit
Joinery	$\frac{1}{2}$ or 1 unit
Bookkeeping— Accounting	1 unit
Business Law	$\frac{1}{2}$ unit
Shop	$\frac{1}{2}$ or 1 unit
Machine Shop	$\frac{1}{2}$ or 1 unit
Pattern Making	$\frac{1}{2}$ or 1 unit
Commercial Geography	1 unit
History of Commerce	$\frac{1}{2}$ unit

Any other credited High School subject, 1 unit.

A unit, as defined by the College Entrance Examination Board, represents a year's study in any subject in a secondary school, constituting approximately a quarter of a full year's work. As a minimum this represents one hundred and twenty sixty-minute hours or their equivalent.

A four-year secondary school curriculum should be regarded as representing not more than sixteen units of work.

The classification for special students is provided in order to make available the privileges of the College to mature men to whom technical instruction in a limited field would be a benefit, but who have neither the need nor the necessary time for a full course of study. Such students are expected to conform to the same standards of attendance and scholarship as are required of matriculated students. Special students may receive a statement of work completed.

Students who wish to transfer from other institutions must present complete credentials including a statement of honorable dismissal to the Registrar. Applications for admission by transfer students will be considered only if the applicant meets all requirements which govern the admission of Freshmen.

A student must satisfy the College of adequate preparation. To do this he may offer either:

(A) Any one of the following examinations covering the subjects required for admission:

1. Those given by the College Entrance Examination Board. ^[8]
2. The entrance examinations given by the College.
3. The examinations given by the Education Departments of the States of New York and New Jersey to students who have satisfactorily completed the courses in the High Schools;

or

(B) A certificate of graduation from an approved High School showing the time devoted to each subject and the standard attained. Entrance credits will be allowed for those subjects only which are satisfactorily credited on the certificate.

^[7] For commercial and vocational subjects, not more than four units, provided that not more than one unit be offered in any one subject.

^[8] For information regarding these examinations see the next following pages.

COLLEGE ENTRANCE EXAMINATION BOARD

Scholarship Examinations and Early Examinations for Admission to Be Held on April 15, 1939

The College Entrance Examination Board will hold examinations at approximately 150 places in the United States on April 15, 1939, for applicants for scholarships and for admission to college. A list of these places will be published about December 15. A detailed announcement regarding the April series of examinations will be ready for distribution on October 1.

Every candidate is required to file an application with the Executive Secretary of the College Entrance Examination Board, 431 West 117 Street, New York, N. Y., not later than March 25, 1939. A candidate who takes the examinations for admission to college should secure a blank form of application for examination from the College Entrance Examination Board; a candidate who takes them in order to qualify for a scholarship must secure the blank form from the college offering the scholarship. In order to facilitate the making of arrangements for the proper conduct of the examinations, it is desired that all applications be filed as early as possible. The examination fee of ten dollars should accompany the application for the scholarship tests^[9] and should be remitted by postal order, express order, or draft on New York to the order of the College Entrance Examination Board.

Applications for examination will be accepted after March 25, 1939, when it is possible to arrange for the examination of the candidates concerned, but only upon payment of five dollars in addition to the regular fee.

When a candidate has failed to obtain the required blank form of application, the regular examination fee will be accepted if it arrives not later than March 25, 1939, and if it be accompanied by a memorandum with the name and address of the candidate, the center at which he will report for examination, the college to which his report is to be sent, and the series of examinations (admission or scholarship) he wishes to take.

No candidate will be admitted to a test late, that is, after the test has begun. Each candidate who is registered for the Scholastic Aptitude Test will receive a booklet containing a specimen test with blank spaces to be filled

in by the candidate. In order to secure admission to the test, the candidate must present not only his ticket of admission but also this booklet with the spaces filled in as requested.

Inasmuch as no special preparation will be needed for this series of examinations, detailed information regarding them, with the exception of the practice booklet for the Scholastic Aptitude Test, will not be distributed to candidates.

The College Entrance Examination Board will report to the institution indicated on the candidate's application the results of his examinations. Candidates should not expect to receive from the Board reports upon their examinations.

[9] The fee for the admission series this year will be five dollars. This series will consist of a form of the Scholastic Aptitude Test containing a verbal and a mathematical section. There will be no separate foreign language and mathematics tests.

COLLEGE ENTRANCE EXAMINATION BOARD

Examinations of June 17-24, 1939

The College Entrance Examination Board will hold examinations in June 1939 at more than three hundred points in this country and abroad. A list of these places will be published about March 1, 1939. Requests that the examinations be held at particular points should be transmitted to the Executive Secretary of the College Entrance Examination Board not later than February 1, 1939.

Detailed definitions of the requirements in all examination subjects are given in a circular of information published annually about December 1. Upon request to the Secretary of the College Entrance Examination Board a single copy of this document will be sent to any teacher without charge. In general, there will be a charge of thirty cents, which may be remitted in postage.

All candidates wishing to take these examinations should make application by mail to the Secretary of the College Entrance Examination Board, 431 West 117th Street, New York, N. Y. Blank forms for this purpose will be

mailed by the Secretary of the Board to any teacher or candidate upon request by mail.

The applications and fees of all candidates who wish to take the examinations in June 1939 should reach the Secretary of the Board not later than the dates specified in the following schedule:

For examination centers

In the United States east of the Mississippi River or on the Mississippi	May 29, 1939
In the United States west of the Mississippi River or in Canada	May 22, 1939
Outside of the United States and Canada, except in Asia	May 8, 1939
In China or elsewhere in the Orient	April 24, 1939

An application which reaches the Secretary later than the scheduled date will be accepted only upon payment of \$5 in addition to the regular examination fee of \$10.

When a candidate has failed to obtain the required blank form of application, the regular examination fee will be accepted if the fee arrive not later than the date specified above and if it be accompanied by a memorandum with the name and address of the candidate, the exact examination center selected, and a list of the subjects in which the candidate is to take the Board examinations.

When the examination supplies of the local supervisor permit, candidates who have failed to file an application with the Secretary may be admitted, upon payment to the local supervisor of a fee of five dollars in addition to the regular fee, to all examinations except the Scholastic Aptitude Test. Such candidates should present themselves at the beginning of the period of registration. A candidate who registers with the supervisor will receive from him a blank form of application and an identification card which must be filled out and handed to the supervisor for transmission to the Secretary of the Board.

In order to exhibit their tickets of admission, to present their identification cards, and to obtain seats in the examination room, candidates should report for a morning examination at 8:45 and for an afternoon examination at 1:45.

An examination will close for candidates admitted late at the same time as for other candidates. The examinations will be held in accordance with the time (Standard Time or Daylight Saving Time) observed in the local schools.

No candidate will be admitted to the Scholastic Aptitude Test late, that is, after the test has begun.

The Scholastic Aptitude Test may be taken upon the completion of the school course or at the end of the third year of secondary school work. Each candidate desiring to take this test, even though he is to take no other examination, must file with the Secretary of the Board the usual application for examination. Every candidate who registers for the test will receive a practice booklet containing a specimen test with blank spaces to be filled in by the candidate. In order to secure admission to the test, the candidate must present not only his ticket of admission but also this booklet with the spaces filled in as requested. If the Scholastic Aptitude Test is taken in connection with other examinations, no additional fee is required; if taken alone, the fee is \$10.

EXPENSES

Tuition

To students entering the College for the first time in September of 1939, and later, the tuition will be ninety dollars (\$90.00) per semester for residents of the State, and one hundred and eighty dollars (\$180.00) per semester for non-residents.

Students who entered the College in February of 1939, or earlier, will be charged a tuition fee of eighty-five dollars (\$85.00) per semester, including registration fee, if residing in the State, and one hundred and seventy dollars (\$170.00) per semester if non-residents.

Students who leave the institution before any semester is completed are *not* allowed rebate of tuition for the remaining portion of the semester.

Students who withdraw before completing the first semester must obtain the Dean's approval in order to avoid being charged for second semester tuition.

Tuition charges are somewhat higher for special programs of study differing from those shown in this bulletin.

Fees

All students entering the College for the first time as candidates for a degree will be charged a matriculation fee of \$10.00.

All students entering the College for the first time in September of 1939, and later, are required to pay a registration fee of \$5.00 each semester.

A fee of \$1.00 per year is charged for the use of a locker.

An annual fee of \$10.00 is required of all students to cover cost of expendables in connection with laboratory work, and charges resulting from breakage, loss of college property and library fines. In some professional

courses where the cost of expendables is high, the charges may exceed this amount. Liability for these charges is not limited to the stated fee.

Registration is required for each term. Freshmen should be guided by the instructions which are found under "Admission to the College". Other students should register not later than noon of the Saturday before the beginning of the term. An extra registration fee of \$5.00 will be required of those who register late.

A fee of \$2.00 is charged for the removal of a "condition" grade; a fee of \$1.00 is charged for the removal of an "incomplete" grade. These fees are payable although such grades are removed without re-examinations.

For special examinations, taken at times other than those regularly scheduled, a fee of \$2.00 will be charged.

A diploma fee of \$10.00 will be required of all candidates for the Bachelor's Degree; and a fee of \$25.00 for the Professional Degree.

In cases requiring special tests in connection with admission to the College a fee of \$5.00 will be charged.

Students who expect to earn their expenses in whole or in part by outside work should not undertake a full course. They should discuss their plans with the Dean with the object of making up a schedule of studies to fit their particular cases.

This college is primarily one for students who commute between their homes and the school. Dormitories are not provided. Out-of-town students who wish to room in Newark may apply to the College for aid in locating desirable accommodations.

Students are advised to defer expenditures for textbooks until they are certain that changes are not contemplated or necessary.

The College is not responsible for loss of property by fire or theft in its buildings and grounds.

SCHOLARSHIPS

For young men of ability who need financial assistance there are in all twenty-five scholarships available.

Application should be made for these scholarships to the President of the College not later than September the first.

William F. Hoffman Scholarships

Boy Scout Scholarships

In addition to these general scholarships the Board of Trustees of the College have granted two scholarships, the recipients to be chosen by the Newark Council of the Boy Scouts of America.

Henry J. Ruesch Scholarships

In memory of Henry J. Ruesch two scholarships have been donated, the recipients to be graduates of the Newark Technical School.

Herbert P. Gleason Scholarships

By the will of Herbert P. Gleason seven scholarships are available for young men of character and ability.

STUDENT LIFE

The aim of the College is to train and educate earnest and industrious students along technical lines. This is the first and the supreme duty of the College; all other activities are secondary.

The professional departments expect student participation in the activities of the student branches of the national professional engineering societies.

The Faculty recognizes the importance of social and athletic activities if properly co-ordinated with the more serious work of instruction.

ATHLETICS

In common with some of the better technical institutions in the country the College does not support a varsity football or baseball team. The athletic activities are designed to interest the average student and to develop *him* rather than to overdevelop a small group.

Varsity competition in basketball, fencing, tennis and track athletics is encouraged, and the College each year puts teams into the field.

DEPARTMENT OF CIVIL ENGINEERING

PROFESSOR H. N. CUMMINGS
ASSOC. PROF. W. S. LA LONDE
ASST. PROF. O. P. L. ALBERT
ASST. PROF. J. M. ROBBINS
ASST. PROF. R. W. VAN HOUTEN
MR. F. A. BUSSE
MR. D. C. FROST
MR. W. J. HOFFMAN

The Civil Engineering course is planned to prepare young men for municipal engineering work, highway work, construction work, or work in the general field of scientific management. Because of the temporary slowing down of work in the civil engineering field, during recent years, streets, highways, water supply and sewage disposal systems, bridges, etc. have become seriously run-down or inadequate. Trained young men will be needed in the near future to take part in the repair, replacement, or enlargement of these necessities. The trend toward large sectional projects, sponsored by the federal government, indicates another field of opportunity for young men trained in civil engineering.

To give students training in the fundamentals of civil engineering, so that they may qualify for employment in any of the lines of work above mentioned, a problem in municipal engineering is used, providing much of the instruction of Junior and Senior years. A topographical survey and map are made of a tract of land containing about a hundred acres. This tract is then subdivided into streets and lots as in a suburban community. The water supply and sewage disposal systems for this tract are designed in connection with the sanitation course. Streets and roads are located and designed as a problem in the highway course. Bridges, culverts, retaining-walls, etc., such as would be required in municipal, or highway work, are designed as part of the work of the senior course in structures.

Active membership in the Student Chapter of the American Society of Civil Engineers is expected of students in this department.

Field Equipment in Civil Engineering

The department has complete and up-to-date equipment for conducting plane, topographic and hydrographic surveys. The equipment includes plane and engineer's transits, levels, plane tables, current meter, sextants, tapes, level and stadia rods, range poles and all other necessary surveying accessories.

Surveying Summer Camp

The complete facilities of the Summer Camp of the Massachusetts Institute of Technology at East Machias, Maine, are available to a limited number of students who are selected on the basis of proficiency in Surveying and general excellence in scholarship. Applicants will be chosen from the group invited to participate in the Honors Option, and will be given the summer camp assignment in place of the industrial assignment given to the other members of the group. The Newark College of Engineering will give to students receiving this assignment scholarships covering a portion of the camp fee.

The M. I. T. Summer Camp is located in a region particularly well suited to instruction in large scale surveying projects, such as geodetic triangulation, mapping of large areas, reconnaissance for highway or railroad location, hydrographic surveying, stream gaging, etc. The camp itself is equipped with modern buildings, sanitary water supply and sewage disposal, fire hydrants and fire-fighting apparatus, and an emergency hospital. The staff includes a practising physician. Students and faculty live in dormitories (known at the camp as barracks) and eat in the camp dining room. The camp is within a few minutes, by automobile, from the East Machias railroad station. Students attending in 1939 will report during the last week of July and remain at the camp seven weeks.

CIVIL ENGINEERING COURSE, B.S. (C.E.)

Courses offered to students entering September 1937 and later.

FIRST YEAR (Given in 1939-1940)

First Semester

SUBJECT		Attendance Hours	
		Rec.	Lab.
CH	11 Chemistry	3	3
ME	1 Engineering Drawing	1	3
Eng	10 English	4	0
Eng	50 History of Industrial Civilization	1	0
Ind E	11 Principles of Engineering (The College)	2	0
Math	1 Mathematics	3	3
Phys	1 Introductory Problems in Physics	0	3

Second Semester

CH	11 Chemistry	3	3
ME	1 Engineering Drawing	1	3
Eng	10 English	4	0
Eng	50 History of Industrial Civilization	1	0
Ind E	11 Principles of Engineering (The College)	2	0

Math	1 Mathematics	2	2
Phys	1 Introductory Problems in Physics	0	2
Phys	2 Physics	2	3

SECOND YEAR (Given in 1939-1940)

First Semester

CE	1 Surveying	3	3
CH	21 Qualitative Analysis	1	3
ME	2 Engineering Drawing	0	2
Eng	20 English	3	0
Eng	60 History of Industrial Civilization	1	0
Ind E	12 Principles of Engineering (The Industry)	0	2
Math	21 Calculus	3	2
Mech	20 Mechanics	2	0
Phys	3 Physics	4	0

Second Semester

CE	1 Surveying	2	3
CH	21 Qualitative Analysis	1	3
ME	2 Engineering Drawing	0	2
Eng	20 English	3	0
Eng	60 History of Industrial Civilization	1	0
Ind E	12 Principles of Engineering (The Industry)	0	2

Math	21	Calculus	3	2
Mech	20	Mechanics	2	0
Phys	3	Physics	2	3

SUMMER WORK

Ind E	60	Co-operative Work (required for Honors Option Group).		
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THIRD YEAR (Given in 1939-1940)

First Semester

CE	2	Surveying	2	6
CE	10	Sanitation	2	0
CE	20	Highways	3	0
CE	40	Hydraulics	3	3
Ind E	13	Staff Control	0	2
Math ^[10]	31	Differential Equations	3	0
Mech	22	Mechanics	2	0
Phys	30	Strength of Materials	3	0

Second Semester

CE	2	Surveying	2	6
CE	10	Sanitation	4	0
Ind E	13	Staff Control	2	0
Ind E	31	Economics	3	0

Math ^[11]	32	Vector Analysis	3	0
Mech	22	Mechanics	2	0
Phys	30	Strength of Materials	3	3

SUMMER WORK

Ind E	61	Co-operative Work (required for Honors Option Group)		
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FOURTH YEAR

First Semester

CE	11	Sanitation	2	3
CE	21	Highways	0	4
CE	22	Highway Traffic Control	1	0
CE	30	Structures	3	4
EE	75	Electricity	3	0
ME	31	Thermodynamics	3	0
Ind E	14	Staff Control	1	2
Ind E	22	Industrial Management	3	0

Second Semester

CE	11	Sanitation	2	2
CE	22	Highway Traffic Control	1	0
CE	30	Structures	3	4
EE	75	Electricity	3	0

ME	55 Mechanical Engineering	3	3
Ind E	14 Staff Control	1	2
Ind E	41 Accounting	3	0
Ind E	51 Business Law	1	0

NOTE: Students who wish to reduce the amount of work per semester in Freshman and Sophomore subjects may apply to the Dean for a regular five year schedule.

[\[10\]](#) Math 31 is optional but recommended for students who contemplate graduate work.

[\[11\]](#) Math 32 is optional but recommended for students who contemplate graduate work.

CIVIL ENGINEERING COURSE, B.S. (C.E.)

Offered in academic year 1939-40 to students who completed Junior requirements before September 1939

FOURTH YEAR

First Semester

SUBJECT			Attendance	Hours
			Rec.	Lab.
CE	11-2	Sanitation	2	0
CE	21	Highways	0	4
CE	22	Highway Traffic Control	2	0
CE	30-2	Structures	2	8
ME	55	Mechanical Engineering	3	3
Ind E	14	Staff Control	1	2
Ind E	22	Industrial Management	3	0
Ind E	51	Business Law	1	0
Co-operative Office or Field work				

Second Semester

CE	11-2	Sanitation	2	0
CE	21	Highways	0	4
CE	22	Highway Traffic Control	2	0

CE	30-2	Structures	2	8
ME	55	Mechanical Engineering	3	3
Ind E	14	Staff Control	1	2
Ind E	22	Industrial Management	3	0
Ind E	51	Business Law	1	0
Co-operative Office or Field work				

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF CIVIL ENGINEERING

CE 1 SURVEYING: Prerequisite, Satisfactory Sophomore Standing.

A course designed to equip the student with a knowledge of the principles and practice of elementary surveying and closely allied sciences and to enable him to apply this scientific information to the professional work of the civil engineer. For descriptive purposes the course is subdivided as follows:

SURVEYING. A study of plane and topographic surveying, consisting of classwork, fieldwork and drafting. The classwork covers a thorough drill in the principles of these branches of surveying. This is supplemented by field exercises covering the use, care, and adjustment of instruments, and cadastral and engineering surveys of elementary character. The work in the drafting room consists of problems involving the interpretation and preparation of topographic, construction, and property maps. Texts: A, B, C, D, E.

ENGINEERING GEOLOGY. A study of geologic science, with particular emphasis on the relationship between physiography, geology, topography, water supply, and the design of engineering structures. Laboratory studies of common rocks, rock-forming minerals and of topographic and geologic maps are made in connection with this course. Texts: A, F.

GENERAL ASTRONOMY. A short course in general astronomy designed to broaden the background of the student and to enable him to obtain a better grasp of the work in practical astronomy given during the Junior year. Texts: A, G.

Texts: A. Departmental Manual I, "Professional Work of the Sophomore Year".

B. Breed and Hosmer, "Principles and Practice of Surveying," Vol. I.

- C. Breed and Hosmer, *“Principles and Practice of Surveying,”* Vol. II.
- D. Robbins, *“Problems in Surveying, CE 1”*.
- E. Vega, *“Logarithms”*.
- F. Ries and Watson, *“Elements of Engineering Geology”*.
- G. Jeans, *“The Stars in Their Courses”*.

CE 2 SURVEYING. Prerequisite, CE 1.

A continuation of the work begun in CE 1. The course covers the fields of hydrographic and geodetic surveying and practical astronomy. The route surveying classwork, formerly given in this course, is now covered simultaneously in the course in Highways CE 20 and CE 21. Route survey fieldwork is covered in the fieldwork of the surveying course. The course consists of classwork, fieldwork and drafting. The classwork covers a thorough drill in the principles of hydrographic and geodetic surveying and spherical trigonometry and practical astronomy. This is supplemented by fieldwork covering the use of the stadia, plane table and traverse board in the execution of topographic and engineering surveys, the methods of gaging streams, the geodetic and astronomic work necessary for control surveys, and the execution of preliminary and location route surveys for highways and sewers which are used in the design work of the courses in Highways and Sanitation. The work in the drafting room covers all computations and plotting necessary to complete a topographic map from the field surveys.

- Texts:*
- A. Departmental Manual II, *“Professional Work of the Junior Year”*.
 - B. Breed and Hosmer, *“Principles and Practice of Surveying,”* Vol. II.
 - C. Hosmer, *“Practical Astronomy”*.
 - D. Hosmer, *“Geodesy”*.
 - E. Pickets & Wiley, *“Route Surveying”*.
 - F. Robbins, *“Notes on Spherical Trigonometry”*.
 - G. Robbins, *“Problems in Surveying, CE 2.”*
 - H. American Nautical Almanac, 1939.
 - I. Vega, *“Logarithms”*.

CE 10 SANITATION. Prerequisite, Satisfactory Junior Standing.

A study of the principles of sanitary science and public health subdivided as follows:

HYDROLOGY. A study of the principles of hydrology with particular emphasis on their application to problems of water supply and storm water disposal.

PUBLIC HEALTH. A study of the engineering control of communicable diseases through the proper collection, treatment, and disposal of municipal wastes; the provision of safe water, milk and foods; the control of rodents and insects; the sanitation of public buildings; housing; and industrial hygiene.

WATER SUPPLY. A study of the methods used to investigate the water supply needs of a community; the selection of the required supply and the design of the collection works. Distribution works are taken up in a later course.

Texts: Mead, "Hydrology"; Ehlers and Steel, "Municipal and Rural Sanitation"; Babbitt and Doland, "Water Supply Engineering". Certain reference books from a department list, to be read during the summer preceding the taking of the course.

CE 10, 11-1 SANITATION. Prerequisite, Satisfactory Junior Standing.

A study of the principles of sanitary science and public health subdivided as follows:

PUBLIC HEALTH. A study of the engineering control of communicable diseases through the proper collection, treatment and disposal of municipal wastes; the provision of safe water, milk and foods; the control of rodents and insects; the sanitation of public buildings; housing; and industrial hygiene.

HYDROLOGY. A study of the principles of hydrology with particular emphasis on their application to problems of water supply and storm water disposal. Computations and designs are carried out in connection with the study of the water resources of a particular stream.

WATER SUPPLY. A study of the methods followed by engineers in investigating the water supply needs of a community; the location of the required supply; the determination of the proper means of conveying the

water to the community; and the design and construction of works in connection with water supply development.

SEWERAGE. A study of the design, construction and maintenance of storm water drains and sanitary sewers, accompanied by the design of such drains for a small community and the preparation of cost estimates and specifications for the same.

Texts: Ehlers and Steel, "Municipal and Rural Sanitation"; Mead, "Hydrology"; Babbitt and Doland, "Water Supply Engineering"; Metcalf and Eddy, "Sewerage and Sewage Disposal". Certain reference books from a department list, to be read during the summer preceding the taking of the course.

CE 11-2 SANITATION. Prerequisite, CE 10, CH 21.

A continuation of the study begun in CE 10, 11-1 and covering the design and construction of works for the purification of water and the treatment of sewage.

Texts: Babbitt and Doland, "Water Supply Engineering"; Metcalf and Eddy, "Sewerage and Sewage Disposal".

CE 20 HIGHWAY. Prerequisite, CE 1.

This course in Highways consists of lectures, student reports and problems, covering the following topics: highway location, with special attention to the part reconnaissance surveys and traffic surveys play in determining the proper location for a highway; the design of roads, dealing with the establishing of grade lines, street intersections, curves, cross-sections and grade separations; grading; highway drainage; soil studies, especially the characteristics of subgrade soils, the grouping of subgrade soils and the conclusions to be drawn from soil studies; non-bituminous and bituminous materials for low-cost roads; natural subgrade treatments and untreated surfaces; bituminous surface treatments; road-mixed and plant-mixed bituminous surfaces; bituminous-macadam bituminized cement and cement-bound macadam roads; portland-cement concrete pavements; base courses for pavements; both hot-mix and cold-laid bituminous pavements; maintenance of bituminous pavements; brick and block pavements; the construction and location of sidewalks, curbs, gutters, guard rails and other appurtenances; highway beautification and lighting; estimates, contracts,

and specifications; street cleaning and snow removal; and the location and construction of landing fields and runways for airports.

The field work in Highways is given in connection with the field work in Surveying CE 2.

Texts: Bruce, "Highway Design and Construction"; Pickels and Wiley, "Route Surveying"; Van Houten, "Problems in Highways, C. E. 20".

CE 21 HIGHWAYS. Prerequisites, CE 2, CE 20.

This is a course in Highway Design in which two problems are undertaken. In the first, plans are prepared for the improvement and paving of about one-quarter mile of city streets. Details of sidewalks, curbs, pavements and drainage are taken into account. The field notes for this problem are obtained in the course in Surveying CE 2. In the second problem a paper location of a highway is made from a contour map (similar to that prepared in Surveying CE 2) for about two miles of rural highway. Plans are prepared which subscribe to the practice and standards of the New Jersey State Highway Department. Attention is given to alignment, grade and cost, with a special study made of the earth quantities and placement.

Texts: Bruce, "Highway Design and Construction"; Pickels and Wiley, "Route Surveying".

CE 22 HIGHWAY TRAFFIC CONTROL. Prerequisite CE 20.

A course designed to give the student a comprehensive knowledge of the problems encountered in the field of highway traffic control together with a thorough study of current methods of dealing with those problems.

The course is presented in the form of lectures by the instructor, reports by the students, supplementary reading, and, whenever practicable, field studies and analyses of actual traffic problems. The subjects covered are as follows: purposes of traffic control; accident statistics; accident records as a basis for accident prevention with special attention being paid to the use of spot maps, flow diagrams and collision diagrams in analysing traffic problems; legislation and administration as a means of regulating traffic; examination of applicants for drivers' licenses; through and stop streets; critical approach speeds; traffic control at intersections by traffic beacons, traffic officers and traffic signals; studies of rotary and channellized intersections; highway and railway grade crossing elimination; traffic lanes,

centerline markings; highway lighting; parking; education of all groups from the pre-school child to the adult; law enforcement, especially studying the problem of the drinking driver and the “accident repeater”; and traffic courts and violations bureaus.

REFERENCE MATERIAL: This is composed of all available literature in the field of traffic control. A few of the organizations and institutions whose literature is used are as follows: American Association of State Highway Officials; American Automobile Association; American Road Builders' Association; Bureau of Public Roads; Institute of Traffic Engineers; International Association of Chiefs of Police; Iowa State College; Metropolitan Life Insurance Company; Motor Vehicle Department of New Jersey and numerous other states; National Bureau of Casualty and Surety Underwriters; National Conference on Street and Highway Safety; National Safety Council; New Jersey Traffic Commission; Northwestern University Traffic Safety Institute; Portland Cement Association; Travelers Insurance Company; University of Illinois; University of Michigan; and University of Wisconsin.

CE 30-1 STRUCTURES. Prerequisite, First Semester Phys 30.

This course forms a transition between the previous courses of mechanics (statics) and strength of materials and the course in structures CE 30-2 given to the senior civil students. It treats of a more rounded and complete study of reaction and internal stresses in roof trusses and statically determinate bridges by both analytical and graphical methods. Special emphasis is placed upon the construction and use of influence lines. A short time is devoted to the approximate solution of lateral bracing and portals.

Text: Sutherland and Bowman “Structural Theory”, 2nd Ed.

CE 30-2 STRUCTURES. Prerequisites, CE 30-1, complete course. Phys 30.

The work of this course is divided between a theoretical study of statically indeterminate structures and the design of a variety of small structures. A thorough theoretical study is made of the deflection of beams and trusses and of the methods of least work, slope deflection, moment distribution, and the column analogy. Secondary stresses, space framework and wind stresses in buildings receive their proportion of attention. Throughout the work in theory those structures that are to be later designed and detailed are

used for class problems, thereby making a close tie between the theory and design and eliminating an unnecessary amount of duplication in arithmetical calculations. Problems are given in the design of, and complete preparation of plans for roof trusses, buildings, foundations, abutments, retaining walls, trestles, trusses, girders, and frames of concrete, steel, and wood, with a study of timber, riveted, and welded framing. Highway loadings are used in preference to railroad loadings in order to simplify computations. Particular emphasis is placed upon orderly and complete computations, standard and practical considerations of design and detail, and thoroughness and neatness in drafting. Given to senior students in civil engineering.

Texts: Sutherland and Bowman, "Structural Theory, Second Edition"; Caughey, "Reinforced Concrete"; Fuller & Kerekes, "Analysis & Design of Steel Structures"; A. I. S. C., "Steel Construction Handbook". Certain reference books from a department list to be read during the summer preceding the taking of the course.

CE 30 STRUCTURES.

Courses CE 30-1 and CE 30-2 will be combined in one senior course in 1940-41 and later years.

CE 40 HYDRAULICS. Prerequisites, Math. 21, Mech. 20.

The subject matter is the same as in CE 41. In addition, a laboratory course is given, in which the characteristics of flow are studied for various types of conduits and measuring devices, and also for various degrees of viscosity of the liquid. Given to students in civil engineering.

Texts: Cummings and Widdop, "Elementary Hydraulics"; Laboratory Manual of the Mechanical Engineering Department.

CE 41 HYDRAULICS. Prerequisites, Math. 21, Mech. 20.

This is a text-book and problem course. The subject of hydrostatics is treated briefly, from the point of view of review work in physics and applied mechanics. In hydrokinetics, the energy balances are emphasized as providing means of solving problems in theoretic flow through orifices, pipes, open channels, and over weirs. Constant emphasis is placed on the degree of precision obtainable, in practice, by the use of the available experimentally determined constants to modify theoretical computations to

meet actual conditions. Given to chemical, electrical, and mechanical students.

Text: Cummings and Widdop, "Elementary Hydraulics".

DEPARTMENT OF ELECTRICAL ENGINEERING

PROFESSOR J. C. PEET
PROFESSOR A. A. NIMS
ASST. PROF. S. FISHMAN
ASST. PROF. F. E. MCKONE
ASST. PROF. P. C. SHEDD
MR. J. H. JOHNSTON
MR. W. JORDAN, 3RD
MR. F. A. RUSSELL
MR. C. H. STEPHANS

The problems and techniques associated with the production, delivery, utilization and control of energy in the electrical form are given the inclusive title of Electrical Engineering. Any general preparation for the recognition and analysis of these problems and the mastery of these techniques, in their infinite variety, must, of necessity, emphasize the basic conceptions and principles which are of widest application. Specific applications, sufficient in number and variety to maintain the student's interest and broaden his point of view are, however, useful supplements.

The work of the freshman and sophomore years consists, primarily, of the foundation mathematics, physics, English and mechanical drafting usually given to all engineering students. In addition, a course in the fundamental electrical units and their application to the magnetic, electro-static and electric circuits, is given. This is followed by a general engineering course in d-c and a-c circuits. The classroom work is paralleled by a laboratory course in electrical measurements.

During the upper class years the principles of electrical engineering are applied to many problems; characteristics and operation of direct-current, synchronous, and induction machinery, rectification, wave form analysis, transient phenomena, power plant equipment, transmission and distribution and electro-physical measurements in magnetism, in induction and

capacitance by balance methods and in multi-electrode vacuum tube characteristics.

A student branch of The American Institute of Electrical Engineers holds about eight or ten meetings each year. All upper classmen become members and are expected to attend its meetings. The second year men are invited to join the local society.

Since electrical engineering is closely related to mechanical and chemical engineering much material from these branches is included in the course.

The Electrical Laboratories

The electrical laboratories are located on the first floor of the Laboratory Building. A centrally located stock room houses much of the equipment used for test and measurement.

Electric power for the laboratories is obtained from the Public Service Electric & Gas Company through a 240-volt, 3-phase, 60-cycle alternating current line. By means of transformers, motor-generator sets, synchronous converters, oscillators, rectifiers, and storage batteries, direct or alternating current power of wide range of voltage and frequency is available. This power can be distributed to any part of the laboratories on either two or three wire lines, through a carefully planned distribution system.

Equipment is available for setting up all types of electric circuits, reactive and non-reactive, for either direct or alternating current power, together with the usual voltmeters, ammeters, and wattmeters, required in the measurement of these circuits.

Several examples of each of the fundamental types of generators and motors for both direct and alternating current, as well as the usual transformers and various special types are conveniently arranged for study and complete tests.

Special types of instruments for extreme range of voltage current and power are also available, together with special instruments such as frequency meters, power factor meters, electro-static voltmeters, oscillographs, and bridge networks for resistance, capacitance and inductance.

Provision is also made for extensive study of the fundamental operating characteristics of the vacuum tubes of many kinds which are so widely used in the control of electric power in its various forms.

The following equipment is worthy of special mention:

A General Electric Educational Set consisting of a synchronous machine; a wound-rotor induction machine; a squirrel cage induction machine; and a double-current generator, each with a 15 k v a rating and wound for either 1, 2, 3 or 6 phase operation.

A Westinghouse synchronous motor-generator set, rated at 15 k v a and wound for 1, 2, 3 or 6 phases. One machine is arranged as a cradle dynamometer and equipped for phase shifting.

A General Electric sine-wave generator coupled to a synchronous motor. This machine generates a voltage wave which conforms accurately to standard wave form. It has a capacity of 5 k v a for three phases and has a ring gear mechanism for phase shifting.

A two-unit General Electric motor-generator set consisting of one 5 k v a generator capable of single-, three- or six-phase operation at 110/220 volts. This is coupled to a 5-kw, 250-volt d-c machine.

Three mercury-arc rectifiers complete with switchboards and auxiliaries.

Three General Electric Oscillographs complete with all auxiliary apparatus.

One Westinghouse four-element Oscillograph.

One Westinghouse Osiso.

One Sundt Neobeam Oscilloscope.

One Du Mont Cathode-Ray Oscilloscope.

One Westinghouse Audio Oscillator.

One Western Electric Audio Oscillator.

ELECTRICAL ENGINEERING COURSE, B.S. (E.E.)

Courses offered to students entering September 1937 and later.

FIRST YEAR (Given in 1939-1940)

First Semester

SUBJECT		Attendance	Hours
		Rec.	Lab.
CH	11 Chemistry	3	3
ME	1 Engineering Drawing	1	3
Eng	10 English	4	0
Eng	50 History of Industrial Civilization	1	0
Ind E	11 Principles of Engineering (The College)	2	0
Math	1 Mathematics	3	3
Phys	1 Introductory Problems in Physics	0	3

Second Semester

CH	11 Chemistry	3	3
ME	1 Engineering Drawing	1	3
Eng	10 English	4	0
Eng	50 History of Industrial Civilization	1	0

Ind E	11 Principles of Engineering (The College)	2	0
Math	1 Mathematics	2	2
Phys	1 Introductory Problems in Physics	0	2
Phys	2 Physics	2	3

SECOND YEAR(Given in 1939-1940)

First Semester

EE	21 Electricity	5	3
ME	2 Engineering Drawing	0	2
Eng	20 English	3	0
Eng	60 History of Industrial Civilization	1	0
Ind E	12 Principles of Engineering (The Industry)	0	2
Math	21 Calculus	3	2
Mech	20 Mechanics	2	0
Phys	3 Physics	4	0

Second Semester

EE	22 Electric Circuits	6	3
ME	2 Engineering Drawing	0	2
Eng	20 English	3	0
Eng	60 History of Industrial Civilization	1	0
Ind E	12 Principles of Engineering (The Industry)	0	2

Math	21	Calculus	3	2
Mech	20	Mechanics	2	0
Phys	3	Physics	2	3

SUMMER WORK

Ind E	60	Co-operative Work (required for Honors Option Group).		
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THIRD YEAR (Given in 1939-1940)

First Semester

CH	21	Qualitative Analysis	1	3
EE	31	Electric Networks	2	0
EE	33	Electric Machinery	2	3
EE	35	Electron Tubes	1	3
ME	31	Thermodynamics	3	0
Ind E	13	Staff Control	0	2
Math ^[12]	31	Differential Equations	3	0
Mech	21	Mechanics	2	0
Phys	30	Strength of Materials	3	3

Second Semester

CH	21	Qualitative Analysis	1	3
EE	32	Electric Transients	2	0
EE	33	Electric Machinery	3	3

EE	35	Electron Tubes	1	3
Ind E	13	Staff Control	2	0
Ind E	31	Economics	3	0
Math ^[13]	32	Vector Analysis	3	0
Mech	21	Mechanics	2	0
Phys	30	Strength of Materials	3	0

SUMMER WORK

Ind E	61	Co-operative Work (required for Honors Option Group).		
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FOURTH YEAR

First Semester

CE	41	Hydraulics	3	0
EE	41	Electric Transmission Equipment	2	0
EE	43	Electric Machinery	2	3
EE	45	Electrical Measurements	2	3
EE	47	Electrical Design	0	4
ME	55	Mechanical Engineering	0	3
Ind E	14	Staff Control	1	2
Ind E	22	Industrial Management	3	0
Ind E	51	Business Law	1	0

Second Semester

EE	42	Electric Transmission Circuits	3	0
EE	43	Electric Machinery	2	3
EE	46	Electron Tube Circuits	1	3
EE	47	Electrical Design	0	3
ME	16	Machine Design	3	0
ME	55	Mechanical Engineering	3	0
Ind E	14	Staff Control	1	2
Ind E	41	Accounting	3	0

NOTE: Students who wish to reduce the amount of work per semester in Freshman and Sophomore subjects may apply to the Dean for a regular five year schedule.

[\[12\]](#) Math 31 is optional but recommended for students who contemplate graduate work.

[\[13\]](#) Math 32 is optional but recommended for students who contemplate graduate work.

ELECTRICAL ENGINEERING COURSE, B.S. (E.E.)

Offered in academic year 1939-40 to students who completed Junior requirements before September 1939

FOURTH YEAR

First Semester

SUBJECT			Attendance Hours	
			Rec.	Lab.
EE	41	Electric Transmission Equipment	5	0
EE	45	Electrical Measurements	3	6
EE	47-2	Electrical Design II	0	4
ME	55	Mechanical Engineering	3	3
Ind E	14	Staff Control	1	2
Ind E	22	Industrial Management	3	0
Ind E	51	Business Law	1	0
Co-operative Industrial Work.				

Second Semester

EE	42	Electric Transmission Circuits	5	0
EE	46	Electron Tube Circuits	3	6
EE	47-2	Electrical Design II	0	4

ME	55	Mechanical Engineering	3	3
Ind E	14	Staff Control	1	2
Ind E	22	Industrial Management	3	0
Ind E	51	Business Law	1	0
		Co-operative Industrial Work.		

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF ELECTRICAL ENGINEERING

EE 21 ELECTRICITY.

This is the fundamental electrical course for all electrical engineering students. The lecture, class and laboratory method is used. The subject is treated from the point of view of the physicist. The electron theory is the basis. For each phenomenon considered a physical explanation is given as well as a mathematical expression. Particular attention is given to the proper definition of the units of measurement. The relation between these quantities is emphasized by problem work.

Laboratory work in measurements and the proper use of instruments is carried on at the same time.

Texts: Zeleny, "Elements of Electrical Engineering"; Peet, "Laboratory Manual in Electricity".

EE 22 ELECTRIC CIRCUITS.

This is a lecture, recitation and laboratory course in the fundamental electrical units and their proper application to the usual direct current and alternating current circuits. It is given to all electrical students in the second semester of the sophomore year.

A general list of the topics is as follows: Magnetism, electro-magnetism; electric current, pressure and resistance; electric power and energy; series, parallel and series parallel circuits; Kirchhoff's law; three-wire system; electro-statics; dielectric circuit; alternating current circuits containing resistance, inductive reactance and capacitive reactance in series and parallel combinations by graphical, analytical and complex quantity methods; single and polyphase alternating current circuits.

Laboratory work in direct-current and constant frequency alternating current circuits supplements the classroom work.

Texts: Dawes, “Electrical Engineering”, Vol. I & II; Peet, “Laboratory Manual in Electricity”.

EE 31 ELECTRIC NETWORKS.

For the purpose of making analyses of electric networks the following principles are introduced:

Algebra of Complex Quantities

Kirchhoff’s Laws

“T” to “Pi” transformation

Thevenin’s Theorem

Superposition Theorem

Magnetic and dielectric coupling

Resonance

Text: Everitt, “Communication Engineering”; Fishman, “Electric Circuit Projects”.

EE 32 ELECTRIC TRANSIENTS.

The transient conditions existing in direct and alternating current circuits whenever the current values are suddenly changed are of great importance in many electrical problems. To investigate these an analysis is made of the time variation of energy, power, current and voltage whenever there is a readjustment of energy in these circuits. The analysis is made for circuits in which energy is stored in either magnetic or dielectric form or in both forms.

Text: Fishman, “Electric Circuit Projects”.

EE 33 ELECTRIC MACHINERY. Prerequisites, EE 21, EE 22.

The subject matter of this course is presented in three divisions as follows:

- a. A study of direct-current generator and motor characteristic curves, separation of losses, regulation and efficiency, parallel operation, three-wire and pump-back tests, armature reaction and commutation.

- b. A similar study of alternating current machinery includes transformers, alternators, synchronous and asynchronous polyphase motors, single-phase motors and converters.
- c. Armature windings: development from elementary coil of both ring and drum types; representation of both open and closed circuit winding by circular diagrams, tables and vector diagrams; bipolar and multipolar, simplex and multiplex windings; phase distribution and distribution factors.

An extensive laboratory course supplements the class work.

Texts: Dawes, "Electrical Engineering", Vol. I & II; "Standard Handbook for Electrical Engineers" (Sixth Edition); Nims, "Armature Winding Notes", Fishman and Shedd Laboratory Manual "Electric Machinery".

EE 35 ELECTRON TUBES. Prerequisites, EE 21, EE 22.

A study is made of the following electronic devices:

Volt-ampere characteristics of contact rectifiers.

Illumination-response of photo-sensitive devices (conductive, voltaic, vacuum and gas emissive cells).

Emission from tungsten, thoriated tungsten and oxide-coated cathodes.

Characteristics and coefficients of vacuum diodes, triodes and multigrid tubes.

The electron gun.

Ignition and volt-ampere characteristics of cold cathode, hot cathode, and pool cathode gas and vapor tubes.

Methods of controlling output of gas and vapor tubes by means of grids.

Texts: Fishman, "Electronics Laboratory Projects".

EE 41 ELECTRIC TRANSMISSION EQUIPMENT. Prerequisite, Satisfactory Senior Standing.

The accessory apparatus for the production of electric power, and its transmission and distribution to the consumer is the basis of a seminar type course. Each student investigates an assigned topic in some detail and

discusses his findings for the information of the class. Prime movers, generator excitation and voltage regulation, feeder voltage regulation, station wiring layouts, switchboards and switching gear, reactors, relays and relay systems, line disturbances and line protection, plant economics and energy rates, industrial motor application and control are topics treated.

Text: “Standard Handbook for Electrical Engineers” (Sixth Edition).

EE 42 ELECTRIC TRANSMISSION CIRCUITS. Prerequisites, EE 31, EE 32.

Aspects of electric power transmission which are subject to analytical attack are treated by lecture and computation methods.

The topics included are: Calculation of short lines, low voltage distribution, conductor materials, spacing, corona effect, economic voltage and frequency, hyperbolic functions and the calculation of line constants, calculation of long lines by graphical, approximate and rigorous methods, comparison of methods, synchronous machines for power factor and voltage control, effect of transformers included in circuit regulation.

Text: Woodruff, “Electric Power Transmission and Distribution”; Everitt, “Communication Engineering”.

EE 43 ELECTRIC MACHINERY. Prerequisite EE 33.

A continuation of Electric Machinery EE 33.

EE 45 ELECTRICAL MEASUREMENTS. Prerequisites, EE 31, EE 32.

This is mainly a laboratory course illustrating some of the more advanced problems in electric circuits. In the associated classroom work the principles underlying the laboratory problems are discussed and the quantitative relations emphasized. The projects considered fall under the general heading of Measurement Circuits:

Oscillograph study of direct and alternating transients.

Alternating Bridges.

Non-sinusoidal wave analysis.

Balance methods of current measurements.

Phase sequence indicators and meters.

Problems of power measurement in polyphase circuits requiring voltage and current transformers.

Text: Nims, “Advanced Circuits Measurements Manual”.

EE 46 ELECTRON TUBE CIRCUITS. Prerequisites, EE 35, EE 45.

A continuation of EE 45, dealing principally with the following topics:

Direct measurements of Tube Constants.

Essentials of Amplifier Circuits of all types.

Oscillator and Inverter Circuits.

Rectifier and Relay Circuits.

Texts: Everitt, “Communication Engineering”; Nims, “Advanced Circuits Measurements Manual”.

EE 47 (EE 47-1, 47-2) ELECTRICAL DESIGN. Prerequisites, EE 22, EE 33.

The performance characteristic of electric equipment depend upon the materials of which it is made, and upon the arrangement and dimensions of these parts. In this course this relationship for simple equipment, such as resistors and magnets, is studied mainly by computation methods. Some time is taken here for the layout of machinery locations on the floors of a building, which gives some practice in the use of surveying instruments. The quantitative study of materials and their arrangement in electric machinery is carried further into the windings of direct and alternating current machinery, including transformers if time is available.

Texts: Nims, “Electric Machine Design Notes”; Busse, “Surveying Notes”.

EE 71 ELECTRICITY.

This is the general course adapted to the needs of chemical and mechanical engineering students.

It is a lecture, recitation and problem course in the fundamental units and their application to the electric circuits and machines.

The topics are as follows: Magnetism, induction, electric current, pressure and resistance, electric power and energy; electric, magnetic and electro-static circuits; single and polyphase alternating current circuits by graphical, analytical and complex quantity methods.

Texts: Dawes, “Electrical Engineering”, Vol. I; Dawes, “Electrical Engineering”, Vol. II; McKone, Laboratory Manual “Applied Electricity”.

EE 75 ELECTRICITY.

This is a general survey course adapted to civil engineers. It is not followed by laboratory experience. It treats the fundamental units and their application to electric circuits and electric machinery, both direct and alternating current. It is made as broad as possible for these students who have but limited time to give to the subject.

Text: Timbie, "Elements of Electricity".

EE 81 APPLIED ELECTRICITY. Prerequisite, EE 71.

This is the electric machinery course for mechanical engineering students. It offers experience in the wiring, measurement and operation of the usual direct and alternating current machines. The proper application of these machines in industry is treated.

The classroom work is supplemented by a machine laboratory course.

Texts: Dawes, "Electrical Engineering", Vols. I & II; McKone, Laboratory Manual "Applied Electricity".

EE 83 APPLIED ELECTRICITY. Prerequisite, EE 71.

This is the electric machinery course for chemical engineering students. It is similar to EE 81 except that it requires but one semester and therefore is much abridged.

Texts: Dawes, "Electrical Engineering", Vols. I & II; McKone, Laboratory Manual "Applied Electricity".

DEPARTMENT OF INDUSTRIAL CHEMISTRY

PROFESSOR V. T. STEWART
ASSOC. PROF. J. A. BRADLEY
ASSOC. PROF. P. M. GIESY
ASST. PROF. J. JOFFE
ASST. PROF. A. S. KOHLER
MR. F. W. BAUDER
MR. S. J. BAUM
MR. M. FREDERICK
MR. G. C. KEEFFE
MR. L. Z. POLLARA
MR. S. N. SADOFF
MR. T. J. TULLY
DR. A. S. WILLIAMS

The four year course in Industrial Chemistry is broad in scope and is designed to give the student a thorough background in the fundamental sciences, engineering subjects, and the necessary cultural subjects. It forms an adequate basis for advanced courses of a professional nature in science and engineering and, by further training, in the methods of scientific research.

The earlier part of the course provides the essential foundations of mathematics, physics, and chemistry. Later comes thorough drill in the assimilation, acquirement of facility of application, and blending of the more theoretical instruction of the earlier years.

To give the student a sound grasp of the subject matter, problems of various types form an important part of the curriculum. The laboratory work is almost exclusively quantitative. The student is required to record observations and to express experimental data in an orderly and precise manner.

Incidental to the formal instruction are such matters as the use of library, methods of finding all that is known of a particular product or process, and the writing of reports.

Courses of a cultural nature constitute an important part of the curriculum. Their purpose is to develop in the young technical worker an intelligent approach to contacts with his fellow workers and to responsibilities of a broader social nature. Like all technical men, the chemist may become a business executive, in which event some breadth of vision may be of the utmost importance to him.

The Chemistry Laboratories

These laboratories are located on the top floor of the Laboratory Building, the laboratories being two in number. The large laboratory is devoted to courses in General Chemistry and Qualitative Analysis, and the smaller one to more advanced work in analysis and to organic chemistry. There is also a balance room and a stock room, both easily accessible to the two laboratories.

The department possesses the material equipment necessary for work in inorganic, organic, analytical and industrial chemistry. Fume closets are installed in both laboratories for the proper handling of processes involving harmful fumes. A continuous supply of distilled water is furnished by an enclosed still. A parsons generator supplies hydrogen sulfide for the work in qualitative analysis.

Among the various special pieces of apparatus are the following: A Parr calorimeter, a centrifuge, conductivity apparatus, mechanical grinder, pumps for the production of compressed air and suction, various special pieces of apparatus for distillation and filtration and for the handling of gases.

The Chemical Engineering Laboratory

The Chemical Engineering Laboratory is arranged for the study of unit operations by the students of the chemical course. The apparatus is commercial equipment of modern design, which has been equipped with

numerous meters, gauges and measuring devices for accurately testing the machines. The students make quantitative studies of factory operations and of the problems involved in the design of equipment. The unit operations studied include heat transfer, fluid flow, distillation, evaporation, drying, etc.

A copper, forced-convection, vacuum evaporator, having a capacity of one thousand pounds of water per hour is available. This machine is so equipped that accurate heat and material balances may be obtained and operation costs calculated as in commercial practice.

A very flexible apparatus is the copper experimental distillation unit, which is 23 feet high. This consists of a 50-gallon kettle, a 19-plate rectifying column, condensers, cooler and an automatic decanter and feed control. Five of the sections of the column are of Pyrex glass, so the actual operation can be carefully studied. It can be operated on several modifications of batch, steam or continuous distillation.

The batch drier, equipped with automatic temperature and humidity control, is arranged for the study of the drying process as it is carried out in commercial practice.

A rotary suction drum filter, Sweetland, plate and frame, and stoneware suction filter are used in experiments on filtration.

Experiments on grinding include a study of a Ball Mill, Jaw Crusher, Micro pulverizer, and an Attrition Mill.

The laboratory is also equipped with a centrifuge, sulfonator, vacuum crystallizer, autoclave and other pieces of equipment, such as pyrometers, necessary for experimental runs on the various units.

INDUSTRIAL CHEMISTRY COURSE, B.S. (CH.)

Courses offered to students entering September 1937 and later.

FIRST YEAR (Given in 1939-1940)

First Semester

SUBJECT		Attendance	Hours
		Rec.	Lab.
CH	11 Chemistry	3	3
ME	1 Engineering Drawing	1	3
Eng	10 English	4	0
Eng	50 History of Industrial Civilization	1	0
Ind E	11 Principles of Engineering (The College)	2	0
Math	1 Mathematics	3	3
Phys	1 Introductory Problems in Physics	0	3

Second Semester

CH	11 Chemistry	3	3
ME	1 Engineering Drawing	1	3
Eng	10 English	4	0
Eng	50 History of Industrial Civilization	1	0

Ind E	11 Principles of Engineering (The College)	2	0
Math	1 Mathematics	2	2
Phys	1 Introductory Problems in Physics	0	2
Phys	2 Physics	2	3

SECOND YEAR (Given in 1939-1940)

First Semester

CH	21 Qualitative Analysis	1	3
CH	22 Inorganic Chemistry	2	0
EE	71 Electricity	3	0
ME	2 Engineering Drawing	0	2
Eng	20 English	3	0
Eng	60 History of Industrial Civilization	1	0
Ind E	12 Principles of Engineering (The Industry)	0	2
Math	21 Calculus	3	2
Phys	3 Physics	4	0

Second Semester

CH	21 Qualitative Analysis	1	3
CH	22 Inorganic Chemistry	2	0
EE	71 Electricity	3	0
ME	2 Engineering Drawing	0	2
Eng	20 English	3	0

Eng	60	History of Industrial Civilization	1	0
Ind E	12	Principles of Engineering (The Industry)	0	2
Math	21	Calculus	3	2
Phys	3	Physics	2	3

SUMMER WORK

Ind E	60	Co-operative Work (required for Honors Option Group).		
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THIRD YEAR (Given in 1939-1940)

First Semester

CH	31	Physical Chemistry	2	0
CH	32	Quantitative Analysis	3	6
CH	33	Thermodynamics	2	0
EE	83	Applied Electricity	3	3
Ind E	13	Staff Control	0	2
Math ^[14]	31	Differential Equations	3	0
Mech	24	Mechanics	4	0

Second Semester

CH	31	Physical Chemistry	3	0
CH	32	Quantitative Analysis	2	6
CH	33	Thermodynamics	2	0

Ind E	13	Staff Control	2	0
Ind E	31	Economics	3	0
Math ^[15]	32	Vector Analysis	3	0
Mech	24	Mechanics	4	1

SUMMER WORK

Ind E	61	Co-operative Work (required for Honors Option Group).		
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FOURTH YEAR

First Semester

CH	41	Physical Chemistry	3	0
CH	42	Organic Chemistry	3	3
CH	43	Industrial Chemistry	1	3
CH ^[16]	44	Unit Operations	0	3
ME	55	Mechanical Engineering	3	0
Ind E	14	Staff Control	1	2
Ind E	22	Industrial Management	3	0
Phys	30	Strength of Materials	3	3

Second Semester

CH	41	Physical Chemistry	3	0
CH	42	Organic Chemistry	3	3
CH	43	Industrial Chemistry	1	0

CH ^[16]	44 Unit Operations	0	3
ME	16 Machine Design	3	0
ME	55 Mechanical Engineering	0	3
Ind E	14 Staff Control	1	2
Ind E	41 Accounting	3	0
Ind E	51 Business Law	1	0
Phys	30 Strength of Materials	3	0

NOTE: Students who wish to reduce the amount of work per semester in Freshman and Sophomore subjects may apply to the Dean for a regular five year schedule.

^[14] Math 31 is optional but recommended for students who contemplate graduate work.

^[15] Math 32 is optional but recommended for students who contemplate graduate work.

^[16] CH 44 is optional for Senior students.

CHEMICAL ENGINEERING COURSE, B.S. (CH.E.)

Offered in academic year 1939-40 to students who completed Junior requirements before September 1939

FOURTH YEAR

First Semester

SUBJECT		Attendance Hours	
		Rec.	Lab.
CH	41 Physical Chemistry	3	0
CH	42 Organic Chemistry	4	6
CH	46 Chemical Engineering	2	6
ME	16 Machine Design	3	0
Ind E	14 Staff Control	1	2
Ind E	22 Industrial Management	3	0
Ind E	51 Business Law	1	0

Co-operative Industrial Work.

Second Semester

CH	41 Physical Chemistry	3	0
CH	42 Organic Chemistry	4	6
CH	46 Chemical Engineering	2	6

ME	16	Machine Design	3	0
Ind E	14	Staff Control	1	2
Ind E	22	Industrial Management	3	0
Ind E	51	Business Law	1	0
Co-operative Industrial Work.				

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF INDUSTRIAL CHEMISTRY

CH 11 GENERAL CHEMISTRY.

Descriptive inorganic chemistry, chemical theory, and elementary applied chemistry. Besides a study of the chemistry of the elements and their compounds, the course includes a brief survey of certain of the more important industrial processes, such as the manufacture of the elementary gases, the acids, soda, glass, cement, and metals. The laboratory work is chosen so as to illustrate the current lectures. In order to emphasize the quantitative nature of the science, the student is required to solve a large number of numerical problems based on chemical processes and to do a certain amount of actual quantitative work in the laboratory.

Texts: McPherson and Henderson, "A Course in General Chemistry"; McPherson, Henderson and Evans, "Laboratory Manual in General Chemistry"; Bradley, "Problems in General Chemistry".

CH 21 QUALITATIVE ANALYSIS.

For Chemical Students. This course includes the analysis of numerous unknowns for both the anions and the cations. Class work covers the practical and theoretical aspects of analysis, including the theory of electrolytes, ionic equilibrium and the law of mass action.

Text: McAlpine and Soule, "Qualitative Chemical Analysis".

For Civil Students. Laboratory work, the same as CH 21. Class work consists of the chemistry of materials used in engineering work, chemical theory and special topics.

Texts: Leighou, "Chemistry of Engineering Materials"; Cornog & Vosburgh, "Introductory Qualitative Analysis".

For Electrical and Mechanical Students. This course is designed to acquaint the students with the methods of analysis and the application of chemical principles to engineering work. Laboratory work consists of

qualitative analysis, and water and fuel analysis. Class work includes the application of chemical theory and such special topics as alloys, fuels, corrosion and the treatment of water for industrial and sanitary purposes.

Texts: Chapin, "Second Year College Chemistry"; Cornog & Vosburgh, "Introductory Qualitative Analysis".

CH 22 ADVANCED INORGANIC CHEMISTRY.

This course undertakes a more thorough treatment of the modern developments of inorganic chemistry than is possible in course CH 11. Such topics as the mass law, vapor pressure, dissociation, velocity of reaction, and kinetic theory are studied in considerable detail. Attention is also given to the recent ideas of the structure of the atom. The course is profusely illustrated by problems.

Texts: Chapin, "Second Year College Chemistry"; Hougen and Watson, "Industrial Chemical Calculations".

CH 31 PHYSICAL CHEMISTRY. Junior Year.

CH 41 PHYSICAL CHEMISTRY. Senior Year.

These two courses form a continuous treatment, the subject matter of which is selected with a more especial view to the needs of students entering the chemical industries than is usual in this subject. The abstract principles of chemistry are developed in such a way as to emphasize their practical importance, and to lead the student to facility and confidence in the application of theoretical knowledge to his everyday work. A large part of the work consists to the solution of problems by the students. The problems are discussed in detail, the aim being to develop the power to use principles, rather than merely to impart factual knowledge of the phenomena. The topics considered in the course are the pressure-volume relations of gases, the properties of solutions related to molal composition, the conduction of electricity in solutions, the ionic theory, the mass-action law applied to the rate and equilibrium of chemical changes, heterogeneous equilibrium from the phase-rule standpoint, thermo-chemistry and thermo-dynamic chemistry. Under the latter topic are considered the free-energy change attending chemical reactions, the maximum work obtainable from them, the effect of temperature on free-energy and a number of applied topics in electro-chemistry such as electro-motive force of voltaic cells, electrode

potentials in relation to the equilibrium of oxidation and reduction reactions, electrolysis in relation to electromotive force and concentration, and gas polarization. Throughout the two courses, the scientific background of the chemical industries is constantly impressed upon the student. Only by constantly applying the principles to concrete problems will the student acquire such a knowledge and the power to use it in new cases.

Text: Getman and Daniels, "Outline of Theoretical Chemistry" (for CH 31 and CH 41).

CH 32 QUANTITATIVE ANALYSIS.

This course includes both the theory and the practice of quantitative analysis. In the laboratory, training is given in the correct technique of quantitative work, first in volumetric and then in gravimetric determinations. Later, extended analyses are carried out. In the classroom the principles underlying the laboratory work are studied; additional methods also are considered. Throughout the year the student receives training in the calculations of analytical chemistry, with special attention to the errors of measurement. Particular consideration is given to the accuracy of the methods studied, in connection with the requirements of their use, and to the saving of time by proper planning of work and choice of method.

Texts: Kolthoff and Sandell, "Textbook of Quantitative Inorganic Analysis"; Giesy, "Problems in Quantitative Analysis".

CH 33 THERMODYNAMICS.

This follows the lines of course ME 31 Thermodynamics but is adapted to the needs of industrial chemists.

Text: Lichty, "Thermodynamics".

CH 41 PHYSICAL CHEMISTRY.

(See CH 31.)

CH 42 ORGANIC CHEMISTRY. Prerequisites, CH 11, CH 31, CH 32.

This is a course in the principles and practices of organic chemistry. In the lectures and recitations a systematic study of the aliphatic and aromatic compounds is undertaken. In the laboratory certain selected experiments in the analysis and synthesis of organic compounds is carried out as well as

exercises in the study of the chemical properties of various classes of compounds. The course is conducted with special reference to the industrial applications of organic chemistry. Some of the newer physico-chemical tools used in the study of the science will be considered. If time permits, certain special topics such as dye-stuffs, alkaloids, and compounds of biochemical importance will be studied.

Texts: Conant, "The Chemistry of Organic Compounds"; Coghill and Sturtevant, "Organic Compounds".

CH 43 INDUSTRIAL CHEMISTRY.

The class work includes a study of industrial equipment and processes. Safety work in the chemical industry is given particular consideration. The laboratory work covers typical operations and processes of manufacturing chemistry, which are carried out in small scale commercial equipment.

Text: Riegel, "Industrial Chemistry".

CH 44 UNIT OPERATIONS.

This is an introductory course in the unit operations of chemical engineering, the purpose of which is to introduce the student to the use of large scale chemical equipment.

CH 46 CHEMICAL ENGINEERING.

The class work consists of a study of industrial equipment and processes. Safety work in the chemical industry is given particular emphasis. The laboratory work comprises a study of unit operations and processes. Seminars, which include reports on the laboratory work and articles in the technical literature, are held at regular intervals. A portion of the time is devoted to surveying as applied to the chemical plant. This includes work in measurements for equipment foundations, piping, etc. The laboratory work includes evaporation, drying, filtration, distillation, etc. Particular stress is laid on the quantitative aspects and interpretation of the data.

Texts: Perry, "Chemical Engineers' Handbook"; Walker, Lewis, McAdams & Gilliland, "Principles of Chemical Engineering"; Kohler, "Laboratory Manual in Chemical Engineering"; Busse, "Shop Surveying Notes".

DEPARTMENT OF MECHANICAL ENGINEERING

PROFESSOR F. D. CARVIN
ASSOC. PROF. H. E. WALTER
ASST. PROF. F. J. BURNS
ASST. PROF. D. E. DAVIS
ASST. PROF. P. E. SCHWEIZER
MR. H. R. BOOTH
MR. B. ESKIN
MR. K. A. MACFADYEN
MR. D. F. OLIVER
MR. J. L. POLANER
MR. H. E. PURDY
MR. A. REMINGER, JR.
MR. E. W. WRAITH, JR.
MR. D. E. ZELIFF

The Mechanical Engineer is concerned with the problems of design, construction and operation of machine tools, of the power machinery to operate these tools and of power machinery in general, such as refrigeration, ventilation, automotive, hydraulic and heat transfer machines. He is greatly concerned with the problems of industrial management and public affairs in general. The subject matter covered by this department has been developed with these points in mind.

The first two years of the course develop a background of mathematics, physics, chemistry and English upon which the technical subjects of the last two years depend. An introduction to the technical phase of the work is had in this period through such subjects as engineering drawing, shop work and electricity. A practical view-point may be developed by work in industry during the summers.

In the third year, the student is introduced to the several basic courses in engineering such as thermodynamics, heat power, hydraulics, strength of materials, machine design and electricity. A knowledge of the problems of our social and industrial life is obtained by a study of economics, staff control and business law.

The student's time in the fourth year, is divided between strictly technical work in college and an attempt to co-ordinate with this the cooperative industrial work in industry. In this manner, the practical side of his profession is emphasized. The technical work in school consists of applying the principles developed in the previous year to definite engineering problems. This is done by a study of power plants, turbines, internal combustion engines, heating and ventilation and structural design. In the laboratories, tests are made on various machines to determine their operating characteristics; fuels and lubricating oils are examined; materials are studied both from the machine tool view-point and their heat treatment and metallurgical properties. The economic side of the question is further emphasized by courses in management and staff control.

Students who have satisfactorily completed the first three years of work in mechanical engineering may choose the aeronautical option courses in the fourth year. In this course, some of the more general subjects in mechanical engineering are replaced by more specialized instruction in aeronautics.

Arrangements may be made whereby students of aeronautical subjects, who wish to enroll in the aircraft mechanics course at the Casey Jones School in Newark or in flying courses at the various fields, will receive credit in their co-operative work for such time spent in these courses. This work may be taken during the summer or during the regular co-operative period. The cost of such courses must be carried by the student; it is not included in the regular college fees.

The student branch of the American Society of Mechanical Engineers holds eight to ten meetings a year at the college. All students in the department are expected to attend these meetings.

Engineering Drawing

The aim in Engineering Drawing is to so train all the students of the engineering departments that they will be able to write, to read, and to understand the universal language of Engineering Drawing.

The work is designed as training for engineers, not draftsmen, therefore an attempt has been made to eliminate copy work as such, and to place stress upon an understanding of what is being done and of the reasons for doing it that way.

The ability to make good freehand sketches rapidly, easily and accurately is an asset to any engineer. Much time and attention is, therefore, devoted to sketching. An engineer must, also, be able to inspect a drawing and know whether or not it is well drawn, accurate and complete, so training is given in checking drawings.

The work is so planned that opportunities for the exercise of planning, judgment and initiative are given to each student.

Every student is urged to set for himself a high standard in each of the following items:

1. ACCURACY—accuracy is a necessity.

Exactness, completeness, and fitness are of the utmost importance in the work of an engineer. A working drawing, no matter how pleasing its appearance, is worthless if the dimensions on it are incorrect, if important dimensions and notes are lacking, or if a job completed according to instructions on it will not function properly. Drawings lacking in accuracy are not acceptable.

2. APPEARANCE—a good appearance is a predisposing factor.

The appearance of a drawing depends upon a few simple and easily mastered elements.

Balance: No crowding, no great open spaces.

Proportion: Of letters to views—of views to the size of the drawing.

Line Work: Clear-cut and uniform.

Lettering: Well formed and well proportioned. Poorly lettered drawings are not acceptable.

Cleanliness: Keep hands and tools clean.

3. SPEED—Time is the essence of the contract.

Speed depends upon understanding, planning, mastering of technique and most of all upon concentration.

The Mechanical Engineering Laboratories

The Mechanical Engineering Laboratory is designed to meet the general purposes of testing and studying machines and materials. It consists of several distinct sections devoted to the special phases of experimental engineering. These sections are as follows: steam, internal combustion engines, flow of fluids, hydraulics, fuels and oils, metallurgy and heat treatment, and machine tools.

The steam division of the laboratory consists of a steam boiler, simple, automatic and compound engines, low speed and high speed turbines, jet and surface condensers, feedwater heater, pumps, weighing tanks, meters and calorimeters. Each machine is equipped with apparatus for measuring quantities conforming with the A. S. M. E. test codes.

The internal combustion engine section includes Diesel engines, a gas engine, Diesel and gasoline automotive engines and water and air cooled airplane engines. Each test engine is connected to a dynamometer. Instruments are provided for making the necessary measurements called for in standard testing. Several engines are used for study purposes.

The flow of fluids laboratory makes provisions for the study of the flow characteristics of steam, air and liquids. Instruments used in measuring the flow of fluids are examined and tested. Steam nozzles and orifices are attached to a condenser for steam flow tests. A forced draft fan and a two-stage air compressor provide air for the investigation of the flow in ducts, nozzles, orifices and general air machines. A small wind-tunnel permits the examination of air flow around small models. The loss of head in pipes, fittings and valves can be tested.

The hydraulic division contains centrifugal and reciprocating pumps, an open flow channel, weirs, nozzles, orifices, weighing tanks and meters.

Hydraulic machines are tested. The friction loss in pipes and fittings is examined.

The fuel and oil section of the laboratory is equipped to make the various standard test of these materials. This includes calorimeters, viscosimeters, flash and fire point testers, distillation outfit and flue-gas analysis equipment. A lubarometer is used to test the coefficient of friction of lubricating oils and to study the effects of various bearing metals.

The metallographic and heat treatment laboratories are equipped with gas and electric furnaces, each with temperature measuring instruments, for the melting and heat treatment of metals. Microscopes, both visual and photographic, are used for the examination of metal structures. Grinding and polishing machines are provided for the preparation of specimens.

The machine tool section of the laboratory is equipped with lathes, milling machines, shapers, grinders, gas welding outfit and the usual small tools required for metal cutting. While the major part of the machine shop instruction is given in connection with the co-operative industrial work, this laboratory is used to demonstrate shop practice and machine tool methods in general.

The test work in the Mechanical Engineering Laboratory is designed to familiarize the student with the construction details, features of operation, methods of control and the comparative merits of the various machines. The student is taught to operate these machines in the safest possible manner and test them along the lines adopted by the various professional engineering societies.

MECHANICAL ENGINEERING COURSE, B.S. (M.E.)

Courses offered to students entering September 1937 and later.

FIRST YEAR (Given in 1939-1940)

First Semester

SUBJECT		Attendance	
		Rec.	Hours Lab.
CH	11 Chemistry	3	3
ME	1 Engineering Drawing	1	3
Eng	10 English	4	0
Eng	50 History of Industrial Civilization	1	0
Ind E	11 Principles of Engineering (The College)	2	0
Math	1 Mathematics	3	3
Phys	1 Introductory Problems in Physics	0	3

Second Semester

CH	11 Chemistry	3	3
ME	1 Engineering Drawing	1	3
Eng	10 English	4	0
Eng	50 History of Industrial Civilization	1	0

Ind E	11 Principles of Engineering (The College)	2	0
Math	1 Mathematics	2	2
Phys	1 Introductory Problems in Physics	0	2
Phys	2 Physics	2	3

SECOND YEAR (Given in 1939-1940)

First Semester

CH	21 Qualitative Analysis	1	3
EE	71 Electricity	3	0
ME	2 Engineering Drawing	0	2
ME	7 Shop Practice	0	3
Eng	20 English	3	0
Eng	60 History of Industrial Civilization	1	0
Ind E	12 Principles of Engineering (The Industry)	0	2
Math	21 Calculus	3	2
Mech	20 Mechanics	2	0
Phys	3 Physics	4	0

Second Semester

CH	21 Qualitative Analysis	1	3
EE	71 Electricity	3	0
ME	2 Engineering Drawing	0	2
Eng	20 English	3	0

Eng	60	History of Industrial Civilization	1	0
Ind E	12	Principles of Engineering (The Industry)	0	2
Math	21	Calculus	3	2
Mech	20	Mechanics	2	0
Phys	3	Physics	2	3

SUMMER WORK

Ind E	60	Co-operative Work (required for Honors Option Group).		
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THIRD YEAR (Given in 1939-1940)

First Semester

EE	81	Applied Electricity	3	3
ME	10	Mechanisms	3	0
ME	30	Thermodynamics	4	0
Ind E	13	Staff Control	0	2
Ind E	31	Economics	3	0
Math ^[17]	31	Differential Equations	3	0
Mech	23	Mechanics	2	1
Phys	30	Strength of Materials	3	3

Second Semester

CE	41	Hydraulics	3	0
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EE	81 Applied Electricity	3	3
ME	10 Mechanisms	3	0
ME	34 Heat Power	3	3
Ind E	13 Staff Control	2	0
Math ^[18]	32 Vector Analysis	3	0
Mech	23 Mechanics	2	1
Phys	30 Strength of Materials	3	0

SUMMER WORK

Ind E	61 Co-operative Work (required for Honors Option Group).		
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FOURTH YEAR GENERAL MECHANICAL OPTION

First Semester

ME	14 Machine Design	3	3
ME	20 Physical Metallurgy	3	0
ME	37 Applied Heat Power	3	0
ME	50 Mechanical Laboratory	1	3
Ind E	14 Staff Control	1	2
Ind E	21 Industrial Management	3	0
Ind E	41 Accounting	3	0

Second Semester

ME	14 Machine Design	3	3
ME	18 Graphics & Structural Design	3	3
ME	22 Metallography	0	3
ME	36 Power Plants	3	0
ME	50 Mechanical Laboratory	0	3
Ind E	14 Staff Control	1	2
Ind E	21 Industrial Management	2	0
Ind E	51 Business Law	1	0

NOTE: Students who wish to reduce the amount of work per semester in Freshman and Sophomore subjects may apply to the Dean for a regular five year schedule.

[\[17\]](#) Math 31 is optional but recommended for students who contemplate graduate work.

[\[18\]](#) Math 32 is optional but recommended for students who contemplate graduate work.

FOURTH YEAR AERONAUTICAL OPTION

First Semester

SUBJECT		Attendance Hours	
		Rec.	Lab.
ME	14 Machine Design	3	3
ME	20 Physical Metallurgy	3	0
ME	90 General Aeronautics	3	6
Ind E	14 Staff Control	1	2
Ind E	21 Industrial Management	3	0
Ind E	41 Accounting	3	0

Second Semester

ME	14	Machine Design	3	3
ME	22	Metallography	0	3
ME	91	Airplane Structure	3	6
ME	92	Airplane Engines	2	1
Ind E	14	Staff Control	1	2
Ind E	21	Industrial Management	2	0
Ind E	51	Business Law	1	0

MECHANICAL ENGINEERING COURSE, B.S. (M.E.)

Offered in academic year 1939-40 to students who completed Junior requirements before September 1939

FOURTH YEAR GENERAL MECHANICAL OPTION

First Semester

SUBJECT			Attendance Hours	
			Rec.	Lab.
ME	18	Graphics & Structural Design	3	3
ME	22	Metallography	0	3
ME	36	Power Plants	3	0

ME	37	Applied Heat Power	3	0
ME	50	Mechanical Laboratory	1	6
Ind E	14	Staff Control	1	2
Ind E	21	Industrial Management	5	0
Ind E	51	Business Law	1	0
Co-operative Industrial Work				

Second Semester

ME	18	Graphics & Structural Design	3	3
ME	22	Metallography	0	3
ME	36	Power Plants	3	0
ME	37	Applied Heat Power	3	0
ME	50	Mechanical Laboratory	1	6
Ind E	14	Staff Control	1	2
Ind E	21	Industrial Management	5	0
Ind E	51	Business Law	1	0
Co-operative Industrial Work				

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF MECHANICAL ENGINEERING

ME 1 ENGINEERING DRAWING AND DESCRIPTIVE GEOMETRY.

The objective of this course is the development of the students' ability and judgment in the field of engineering drawing. The work of the course is based upon case studies accompanied by short informal lectures upon modern commercial practice and upon the theory on which such practice is based. Emphasis is placed on speed, accuracy, neatness and on the techniques by which these are obtained.

The course includes study of and practice in lettering, line work, projection, conventions, tracing, sketching, shop standards, the use of notes, the reading of blue prints, the reading of layout drawings, pictorial representation, developments, and checking as applied to commercial engineering drawing.

Text: Freshman Engineering Drawing Notes.

ME 2 ENGINEERING DRAWING. Prerequisite ME 1.

The principles laid down in the previous year's work are applied to a series of problems in structural drafting, concrete work, gear drawing developments, and welded steel work. Such additional work in descriptive geometry as is necessary for each drawing is given as the work develops. An opportunity is given to learn modern drafting practice, the way a drafting room is managed, and to develop the basis for future courses in design and engineering.

Text: Sophomore Engineering Drawing Notes.

ME 7 SHOP PRACTICE.

A course for sophomore mechanical students, given in two parts.

PART 1—SURVEYING AND MACHINE LAYOUT. Three hours a week for half a semester are devoted to laboratory work in machine layout. This includes surveys of existing equipment and exercises in establishing lines and elevations for setting machines and equipment.

PART 2—MACHINE TOOLS. Three hours a week for half a semester are devoted to a laboratory course in machine tool practice. Experiments are carried out in the machine shop and visits are made to industrial shops.

Text: Busse, "Surveying Notes"; "Shop Notes", Turner, "Machine Tool Work".

ME 10 MECHANISMS. Prerequisites Math 21, Mech 20.

This course is essentially one of preparation for the succeeding work in machine design. It includes the study of links, bands and contact motion; of gears and gear teeth, epicyclic trains and cams. The recitations and lectures are supplemented by work in the drafting room where numerous problems are solved graphically.

Text: Schwamb, Merrill and James, "Elements of Mechanisms".

ME 14 MACHINE DESIGN. Prerequisites, ME 2, ME 10 and concurrent with Phys 30.

A course for senior mechanical engineering students. This course continues the work of the previous year in Mechanisms. It is outlined to place emphasis on the strength as well as the motion of machine elements and their final assembly into the complete machine. The theory of the graphic solutions of problems is developed and applied to the analysis of the stress in machines, including the effects of friction. Both theoretical and empirical methods are applied to the design of machines. Its purpose is to instruct students to attack problems in a direct and orderly manner. Three hours of lectures and recitations and three hours of drafting room work per week throughout the year.

Text: Faires, "Design of Machine Elements", Fairman and Cutshall, "Graphic Statics".

ME 16 MACHINE DESIGN. Prerequisite, ME 2 and Concurrent with Phys 30.

A course in design for non-mechanical students. This course is subdivided into two parts. Part one deals with general design and is further divided into

two sub-groups. The first part of this sub-group deals with what is commonly called Mechanisms and the second part deals with subject matter which is usually associated with Machine Design courses which should lead to the ability to proportion parts of machine elements. Part two is for the purpose of making the student acquainted with materials and their characteristics through microscopic examination. In this part of the work the student is required to examine not only steels but non-ferrous materials such as brasses and alloys of aluminum as well. This information coupled with that given in Course Phys 30 should acquaint the student with materials from any points of view and should make him conscious of the important part played by materials in design work.

Text: Hyland and Kommers, "Machine Design".

ME 18 GRAPHICS AND STRUCTURAL DESIGN. Prerequisite, Phys 30.

The theory of graphic statics is developed. It is then applied to beams, columns, roof trusses, cranes, etc. The theory of re-enforced concrete is studied. Design problems on conveyers, foundations, cranes, chimneys and walls are worked out. A classroom and drafting room course for senior mechanical engineering students.

Text: Opdyke and Schweizer, "Graphic and Structural Design Notes".

ME 20 PHYSICAL METALLURGY. Prerequisite, CH 11.

This course deals with the study of metals by means of microscopic examination. The subject is introduced by a discussion which aims to define a metal. This is followed by an explanation of metallic properties and the distinction between metals, non-metals, and metalloids. The standard equilibrium diagrams for binary alloys are then studied and include the liquid to solid and the solid to solid transformations that take place. This is followed by a detailed study of the iron, iron-carbon diagram. The effect on the microscopic structure due to the addition of a third element such as nickel, manganese, chromium, vanadium, tungsten, etc., is then studied.

The equilibrium diagrams of the copper-zinc and aluminum copper are then studied in detail.

In addition to these discussions the student is required to prepare for microscopic examination and to take photo-micrographs of such materials as cast iron, cold rolled steel, carbon steels, standard S. A. E. Steels such as

nickel, nickel-chromium and molybdenum steels, brasses and aluminum alloys, etc.

Text: Van Wert, "Introduction to Physical Metallurgy", Woldman, "Physical Metallurgy", "Metallurgy Laboratory Manual".

ME 22 METALLOGRAPHY. Prerequisite, ME 20.

This course consists of three hours a week of work in the Metallographic Laboratory. The subject matter includes the determination of critical points; calibration of thermo-couples; study of gas and electric heat treating furnaces, and the effect of heat treatment on steels, brasses and alloys of aluminum. The student is required to prepare specimens for microscopic examination to study them under the microscope and photograph them in order to determine whether or not he has secured the required structures in heat treatment. The change in such physical properties as hardness is also studied. This course emphasizes the fact that microscopic examination is a valuable adjunct rather than a purely laboratory procedure.

Text: Woldman, "Physical Metallurgy".

ME 30 THERMODYNAMICS. Prerequisites, Math 21, Phys 3.

A Junior Course for mechanical engineering students. The thermodynamic theory of gases and vapors is studied with respect to both source of energy and the methods of making it available. The subject matter includes a study of energy and its availability; the properties of gases and vapors; energy changes during expansions and compressions; the various ideal cycles for converting heat into work; and the general theory of the flow of fluids. A lecture, recitation and problem course.

Text: Barnard, Ellenwood, Hirshfeld, "Heat Power Engineering", Vol. 1; Keenan and Keys, "Steam Tables".

ME 31 THERMODYNAMICS. Prerequisites, Math 21, Phys 3.

A Junior Course for non-mechanical engineering students. The thermodynamic theory of gases and vapors is studied with respect to both source of energy and the methods of making it available. The subject matter includes the properties of gases and vapors; energy changes during expansions and compressions; and the various ideal cycles for converting heat into work. A recitation and problem course.

Text: Faires, “Elementary Thermodynamics”; Keenan and Keys, “Steam Tables”.

ME 34 HEAT POWER. Prerequisite, ME 30.

This course consists of applying the principles of thermodynamics to heat power problems. The subject matter covered includes combustion, heat transfer, steam engine principles, air compressors, air engines and refrigeration. The laboratory work includes fuel and oil testing and studies and tests of instruments and apparatus used in heat power engineering and the flow of fluids.

Texts: Barnard, Ellenwood, Hirshfeld, “Heat Power Engineering”, Vol. 1, 2, 3; Shoop and Tuve, “Mechanical Engineering Practice”. Department Notes.

ME 36 POWER PLANTS. Prerequisite, ME 34.

A course for senior mechanical engineering students. The subject matter consists of a study of modern practice in steam power plants and heating and ventilation systems. Boilers, feedwater heaters, condensers and other plant auxiliaries are studied. The economics of power generation is discussed. Individual problems are assigned on power plants design and on heating and ventilation systems.

Text: Barnard, Ellenwood, Hirshfeld, “Heat Power Engineering”, Vol. 2, 3; Severns, “Heating, Ventilation and Air Conditioning Fundamentals”.

ME 37 APPLIED HEAT POWER ENGINEERING. Prerequisite, ME 30.

A course for senior mechanical engineering students given in two parts.

PART 1—STEAM TURBINES. Three hours a week for one semester are devoted to a thermodynamic study of turbines. Various types of turbines are examined; nozzle design problems are solved; velocity diagrams for impulse and reaction turbines are developed, and the characteristics and design features of turbines in general are studied.

PART 2—INTERNAL COMBUSTION ENGINES. Three hours a week for one semester are devoted to a study of internal combustion engines, including gas, gasoline and Diesel engines. The various air standard cycles are studied. Ignition, carburetors and injection mechanisms are discussed. The

design features of the various engine parts are considered. Attention is paid to the cooling and lubrication systems.

Text: Barnard, Ellenwood and Hirshfeld, "Heat Power Engineering", Vol. 1 and 2.

ME 50 MECHANICAL LABORATORY. Prerequisite, ME 34.

A course for senior mechanical engineering students. The course consists of one hour of lecture and six hours of work in the experimental laboratory. Studies and tests are conducted on steam engines, turbines, condensers, boilers, gasoline and Diesel engines and hydraulic equipment. Air compressors and refrigeration units are examined and tested; the flow of fluids is studied; oils and fuels are tested.

Texts: Shoop and Tuve, "Mechanical Engineering Practice"; Department Laboratory Manual.

ME 55 MECHANICAL ENGINEERING. Prerequisite, ME 31.

This course in heat power engineering is for students in chemical, civil and electrical engineering. Classroom theory is correlated with practice in the laboratory.

PART 1. HEAT POWER ENGINEERING. Three hours a week of lectures, discussion and problems in applied thermodynamics and heat engines. The course covers study of fuels, combustion, boilers, feedwater, heat transfer steam engines and turbines, pumps, internal combustion engines, air compressors, and refrigeration.

PART 2. MECHANICAL LABORATORY. Three hours a week. The experimental work in the laboratory includes test on steam engines and turbines; gasoline and diesel engines, pumps, and hydraulic equipment; fuel calorimetry and exhaust gas analysis.

Texts: Craig and Anderson, "Steam Power and Internal Combustion Engines"; Shoop and Tuve, "Mechanical Engineering Practice"; Laboratory Notes.

ME 90 GENERAL AERONAUTICS.

Recitations, lectures and trips to airports. Fundamental principles of aeronautics including a study of stability and control of airplanes.

Description of modern aircraft. Air transportation in its engineering, traffic and economic aspects. Airplane maintenance.

ME 91 AIRPLANE CONSTRUCTION.

Lectures, drafting and laboratory work. Analytical and graphical analysis of airplane parts. Materials used in airplane construction. Shop methods of construction.

ME 92 AIRPLANE ENGINES.

Lectures and laboratory work. A study of airplane engines. Thermodynamics of internal combustion engines; mechanical design including power, fuel and carburetion. Laboratory testing of different types of airplane engines.

DEPARTMENT OF ENGLISH

ASSOC. PROF. P. M. GIESY
ASST. PROF. F. A. GRAMMER
ASST. PROF. L. C. SPRY
MR. WILLIAM ARNOTT
MR. F. C. BURT
MR. G. A. VALENTE

The technical work of the engineer requires him to write notes, letters, and reports in a clear, correct, and concise manner. He must be able to read both technical and non-technical writing quickly and accurately. His advancement will depend upon the impression which he makes upon his superiors. Consequently the spoken English which he uses in conversation and in meetings will affect his professional progress. The reading of good literature will help him to understand how different sorts of people will act and feel under various conditions, and so will aid him in solving problems involving personal relations.

As the engineer advances in his profession his contacts are more and more with men who do not have a technical background. With these men he cannot use scientific language, the terminology and formulas of chemistry or mathematics. If he is to make himself clear, he must be able to use the English language in a way which will make his hearers understand his facts and ideas. If he is to persuade his hearers that what he advocates is the proper thing to do, he must speak or write so that they feel that he is a competent, trustworthy man.

During the first two years of his course, the student is trained in writing, speaking, and reading. The training consists largely in practice: in writing on assigned subjects, in speaking before a group of about a hundred, and in reading selections from literature which have interest to an engineer. Practice in writing and speaking continue during the last two years of the course: the student prepares and presents reports in his various professional courses.

Concurrent with the work of the first two years in English is a series of lectures on the history of industrial civilization. These are intended to give the student an appreciation of the broader aspects of engineering development and particularly of its social results. They also furnish material for the written work of the student in his study of English.

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF ENGLISH

Eng 10 ENGLISH.

The aims of this course are to train the student to express ideas in writing and speech, and to read rapidly and accurately. He learns to write by writing, by having his mistakes pointed out, and by correcting them. He learns to speak by speaking before a group of about a hundred, and by being criticized by the group and by the instructor.

- (a) Selections from literature which have engineering interest are read, some of them being abstracted. Translations from foreign classics are included with excerpts from English literature.
- (b) Two hours per week are given to the writing of themes, based in general on the material gathered in (a) and in Course Eng 50. One hour per week is spent in class work on the principles of English composition, particularly as applied to technical writing.
- (c) One hour per week is applied to practice in oral English. At first the students read before the class themes which they have written; later they speak without notes on current topics of engineering interest.

Texts: Giesy and Arnott, "Technical English Composition"; Park, "English Applied in Technical Writing"; Cullimore, "Selections for Engineering Students".

Eng 20 ENGLISH.

The work of course Eng 10 is continued throughout the second year.

- (a) During the summer preceding this course each student is required to read and report on five books of general engineering interest. The reading and abstracting of literature selections is continued.
- (b) Composition work is continued, two hours per week. Some practice is given in the abstracting of technical articles, and in the writing and dictation of business letters.

(c) Practice in speaking continues throughout the year.

Texts: Giesy and Arnott, "Technical English Composition"; Park, "English Applied in Technical Writing"; Cullimore, "Selections for Engineering Students".

Eng 50 HISTORY OF INDUSTRIAL CIVILIZATION.

Lectures on the history of civilization from earliest times to the Industrial Revolution. Particular attention is paid to the developments of science, technology, and industry, to the social influences affecting these developments, and to their social results.

Eng 60 HISTORY OF INDUSTRIAL CIVILIZATION.

A continuation of the lectures of Course Eng 50, through the Industrial Revolution and down to the present. Special consideration is given to technological unemployment and the other social problems connected with the development of labor-saving machinery.

DEPARTMENT OF INDUSTRIAL ENGINEERING

PROFESSOR J. A. BROOKS
ASSOC. PROF. R. WIDDOP
ASST. PROF. C. J. KIERNAN
ASST. PROF. G. D. WILKINSON
MR. P. L. CAMBRELING
MR. J. C. HOFFMAN
MR. O. J. SIZELOVE
MR. R. I. VAIL
MR. J. W. WILLARD

An important function of this department is to test, orient, and guide the student. Starting in the freshman year the student is advised, by means of psychological tests, as to his fitness for engineering. Later, the Staff Control course offers an opportunity to guide the student in the field of human relationships. Guidance in this field is as important as it is in engineering study.

Many engineering graduates are entering the fields of manufacturing, selling, and administration. These men should have not only a knowledge of the fundamentals of engineering, but also a knowledge of economic theory, business functions, and human relationships. It is also believed that a knowledge of these subjects will be beneficial to those men who remain in the engineering field.

Therefore, in addition to the training in the fundamental principles of engineering, every student in the Newark College of Engineering is required to take all of the courses listed in the Department of Industrial Engineering.

This department serves as a link between industry and the college. It is responsible for the direction and administration of student work in industrial plants and organizations. Beginning in the freshman year,

students are interviewed several times to determine their fitness and their preferences regarding placement in industrial plants. Records of these interviews are used in placing students in industrial work and in summer work.

The department arranges for the placement of students in industry and for the details of working plans or programs for each student. After the student is placed, contact with the employer is maintained by frequent visits to the plant. The progress of each student is carefully supervised, and, in cooperation with a representative of the firms, records are kept showing the progress of each student throughout the period of his employment. Students must receive a satisfactory mark in their industrial work before they are eligible for graduation.

The department seeks to give the student effective individual *guidance*, a gradual, consistent *orientation* to his professional life, and a keen awareness of human relation values of his obligations as a citizen and a member of society.

The department uses as mediums, certain courses extending over the college career, established *psychological* tests, supervised industrial placement, personal interviews, thorough treatment of Staff Control, Economics and Management and lectures and group discussions with men prominent in fields connected with this program.

The department also acts as a link between its graduates and industry. It conducts a placement bureau for the purpose of helping the graduates better their employment opportunities.

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF INDUSTRIAL ENGINEERING

Ind E 11 PRINCIPLES OF ENGINEERING. (The College).

An attempt to interpret to the new student the activities which go to make up his college experience.

The various catalogued subjects are discussed and their values indicated and explained as an integral part of the professional development of an engineer. The why and wherefore of the various courses and their objectives are discussed with the students so that they may have a clear and somewhat definite idea as to the reasons for courses and instruction material.

As the course proceeds extra-curricular activities are considered in their relation to the academic values.

The objective of the course is to give some purposeful direction to the activities of the freshman and to possibly establish some sense of values inherent in his undergraduate work.

The course consists of a series of two-hour discussion periods in subjects carefully and sequentially arranged and definitely scheduled. The discussions are led by members chosen from the entire staff for their particular fitness to explain professional values in terms within the students' experience.

Ind E 12 PRINCIPLES OF ENGINEERING (The Industry).

A series of discussions centering about personal problems having for its object the preparation of the student for the required industrial experience to follow his sophomore year. Believing that general attitudes, point of view, emotional and personality factors are of the greatest importance in any common enterprise, the discussions are so directed as to clarify some of the

common misconceptions concerning modern industrial practice as it touches the individual.

A serious attempt is made to induce the student to undertake some constructive program of personality and character development, to meet the standards required of those who can successfully live and work together. While the work of the previous year has served to explain the internal standards and objectives of the college, this continuation of the course explains the aims, objectives and values inherent in the great industrial laboratory.

Ind E 13 STAFF CONTROL.

The subject matter of this course includes those factors which have to do with human behavior and with human relation problems. It deals with the coordination of the young engineer with his environment. Particular attention is given to the professional, economic, social, emotional and moral phases of this correlation.

In the junior year the students discuss twenty-five to thirty books on human relations and subjects which affect human behavior. The subject matter of these books, in outline form, is presented to the class by the students.

Ind E 14 STAFF CONTROL.

In the senior year the students discuss problems in human relations. Many of these problems, which are presented in written form by the students, consist of incidents which have taken place in the cooperative firms. Therefore the cooperative firms may be viewed as human relation laboratories, furnishing real and vital problems for the students.

The discussion groups provide an opportunity for the students to discuss and analyze these problems which now face them; and also problems which may face them later in their profession and their every day life.

Ind E 21 INDUSTRIAL MANAGEMENT.

This course includes a study of the industrial plant, its design, layout, and equipment; organization, production control, time and motion study, standardization, cost finding and engineering economy studies. The object of the course is to present to the students some of the important principles underlying modern management methods.

Texts: Alford, "Cost and Production Handbook"; Grant, "Problems of Engineering Economy".

Ind E 22 INDUSTRIAL MANAGEMENT.

This course is similar to Ind E 21 but is less extensive in nature.

Text: Kimball, "Principles of Industrial Management".

Ind E 31 ECONOMICS.

This course in the fundamentals of Economics is presented from the business man's point of view and includes many concrete examples and illustrations from the world of business. Some of the subjects discussed are economic concepts, the nature of production, organization of modern business, size of business units, specialization, process of exchange, money, business cycles, monopolies, international trade, business risks, distribution of income.

Text: Bye, "Principles of Economics".

Ind E 41 ACCOUNTING.

This course is intended to give the engineering student the fundamentals of accounting. Only enough general bookkeeping and accounting is supplied to make the course practical and to provide a proper groundwork. Emphasis is placed on the preparation and analysis of statements, the calculation of fixed assets, and control features.

Text: Reitell and Van Sickle, "Accounting Principles for Engineers".

Ind E 51 BUSINESS LAW.

An elementary study of the principles of the Common Law as applied to business relations. Students are required to study definite parts of the text in preparation for each meeting of the class. The class time is divided between discussion of the subject matter and written quizzes. The subjects given particular attention are: The definition of "contract", and a detailed study of each element of the definition; agency; sales; partnerships and (briefly) corporations; negotiable instruments; patents, copyrights and trade-marks; master and servant; damages; evidence.

A practicing patent attorney gives special lectures on the subject of patents.

Text: Harding and Canfield, "Legal and Ethical Phases of Engineering".

Ind E 60 COOPERATIVE WORK.

This is industrial placement offered to selected groups of pre-Junior students who have demonstrated their ability, and have given some indication that they will capitalize the experience.

The placement is designed to furnish the laboratory work for the courses in Staff Control, Management, and Economics, and to provide general motivation for the professional courses given during the last two years. The student gains the experience of adjusting himself to a new and usually different environment, learns at first-hand some of the factors which affect a young man's progress in industry, and has the opportunity to observe the practical application, and the limitations, of some of his academic subjects.

The College maintains close contact with the industrial firm and obtains frequent reports on the student's personal qualities. These reports are discussed with the student in individual consultations.

Ind E 61 COOPERATIVE WORK.

This work is similar to Ind E 60. It is offered to selected groups of pre-Senior students.

DEPARTMENT OF MATHEMATICS

PROFESSOR J. H. FITHIAN

ASST. PROF. E. G. BAKER

MR. E. C. EASTON

MR. C. KONOVE

MR. P. MAINARDI

MR. E. M. SQUIRE

Confidence in his ability to solve the mathematical problems which will confront him in training and practice is a very necessary qualification for the successful student of engineering. Not only must he be able to solve these problems, but he must know that his solutions are correct. The courses in mathematics given to all students during the freshman and sophomore years aim to provide this confidence.

Emphasis is placed not on acquiring information, but on developing skill,—skill in analyzing problems and arriving accurately and efficiently at their solution. Consequently, much time is given to written work under careful supervision of the instructors. Neatness and orderly arrangement are stressed, as well as efficiency of methods and the checking of results.

In the sophomore year the same division of time between classroom recitations and written exercises is continued. Here the student adds a powerful tool to his equipment in the theory of the differential and integral calculus. Applications of the methods studied include many practical problems from various types of engineering work.

Beside the minimum requirements for completing the courses mentioned above, a large number of extra problems is included to provide further training for students of superior ability. For those who plan to go into fields of research or to continue their studies after graduation, the Department offers certain advanced courses designed as preparation for graduate work. These may be elected by upper classmen in addition to the regular courses.

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF MATHEMATICS

Math 1 FRESHMAN MATHEMATICS.

In order to enable him to handle accurately and efficiently the mathematics of engineering subjects, the student is given a thorough training in the analysis and solution of problems, and in the performance of numerical calculations, including the use of slide-rule and logarithmic methods.

The following subject matter will be included:

PLANE TRIGONOMETRY: review of the solution of right and oblique triangles and fundamental trigonometric analysis.

GEOMETRY: review of the use of mensuration formulas for plane and solid figures.

ALGEBRA: review of fundamental operations, and simplification of fractional forms; solution of equations and simultaneous equations, linear and quadratic, and the approximate solution of equations of higher degree; exponents and radicals; complex numbers, variation, binomial theorem, and progressions (with applications to compound interest and annuities).

ANALYTIC GEOMETRY: fundamental formulas; general curve plotting; equations of the straight line, circle and conic sections (with applications); polar coordinates; translation and rotation of axes; and an introduction to solid analytic geometry.

Texts: Oglesby and Cooley, "Plane Trigonometry"; Pettit and Luteyn, "College Algebra"; H. B. Phillips, "Analytic Geometry".

Math 21 CALCULUS. Prerequisite, Math 1.

Topics include the technique of differentiation; maxima and minima, rates, curvature, parametric equations, differentials, series, and partial differentiation; technique of integration; areas, volumes, lengths, surfaces, centroids, moments of inertia, fluid pressure, work, multiple integrals, and approximate integration by Simpson's Rule.

The theory and technique of both differentiation and integration are studied during the first term, with a few simple applications, mostly geometric in character. The second term affords opportunity for many practical applications from various fields of engineering. The aim of a set of general review problems during the last few weeks is to teach not only how to use the methods previously studied, but when to use them—i.e., whether the nature of a problem suggests an exact analytical solution, or an approximate or graphical solution.

Texts: Granville-Smith-Longley, “Elements of the Differential and Integral Calculus”; N. C. E. “Laboratory Manual for a Course in Calculus”.

Math 31, 32.

Two advanced courses, Differential Equations (first term) and Vector Analysis (second term), are optional for Juniors in addition to the work of the regular curriculum. No attempt will be made to give an exhaustive mathematical treatment, but certain parts of these subjects will be taught together with other related material necessary for the solution of important problems in all branches of engineering.

Text: Doherty and Keller, “Mathematics of Modern Engineering”. Vol I.

Math 31 DIFFERENTIAL EQUATIONS. First and second order equations of common occurrence; linear differential equations of any order with constant coefficients, and systems of linear equations; determinants; Fourier series and harmonic analysis.

Math 32 VECTOR ANALYSIS. Algebra and calculus of vectors; line and surface integrals, and potential theory; vector operators, and their application to electromagnetic theory and the derivation of certain partial differential equations of mathematical physics.

DEPARTMENT OF MECHANICS

PROFESSOR B. S. KOSHKARIAN

ASST. PROF. J. JOFFE

MR. P. O. HOFFMANN

The courses in mechanics are designed to provide the student with a sound foundation in a subject which occupies a position of basic importance in all branches of engineering and especially in the analysis and design of machines and structures.

While some emphasis is placed on routine calculations and development of formulas, the main objective of the courses is to present general methods of attack and a scientific point of view. The greatest emphasis is placed upon the ability to carry on sustained work at reasonably high levels.

A considerable portion of the time in the courses is devoted to the solution of problems of a practical nature and largely drawn from the field of engineering. In connection with these problems stress is laid on clearness of statement and accuracy of formulation and solution. The technique and methodology are considered of extreme importance in undergraduate study.

The recitations are individual as far as possible and are supplemented by group discussions. It is believed that progressive tests are the fairest criteria for determining the students' mastery of the subject. Written examinations form an essential part of the courses.

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF MECHANICS

Mech 20 STATICS. Prerequisites: Math 1, Phys 1, 2.

The course is designed to provide the prospective engineer with a thorough training in the fundamentals of statics, which form an indispensable background for the study of engineering subjects of a more specialized character. The student is acquainted with the underlying assumptions and broad general principles of the science and is encouraged to apply them in the solution of a great variety of problems of practical interest to the engineer.

The principal topics covered in this course are: composition and resolution of forces and couples; equilibrium; analysis of simple frameworks; flexible cables; the laws of friction with general application and special reference to journal, belt and pivot friction, and rolling resistance.

Texts: Seely and Ensign, "Analytical Mechanics"; Joffe, "Problems in Mechanics".

Mech 21 KINEMATICS AND KINETICS. Prerequisites: Mech 20, Math 21.

This course treats of the laws governing motions of bodies with applications to conditions most frequently met in engineering practice. The principal topics covered under kinematics are: linear and angular displacement, velocity, and acceleration; rectilinear and curvilinear motion; motion curves; relative motion; motion of rigid bodies; instantaneous center. The principal topics covered under kinetics are: Newton's laws applied to the motion of a particle; D'Alembert's principle; motion of the mass-center; translation, rotation and plane motion of a rigid body; work, power, energy, impulse, and momentum; principles of work and energy, principles of impulse and momentum, and their application to special types of motion of rigid bodies.

Texts: Seely and Ensign, "Analytical Mechanics"; Joffe, "Problems in Mechanics".

Mech 22 KINEMATICS AND KINETICS. Prerequisites: Mech 20, Math 21.

The general aim and content of this course is the same as that of Mechanics 21. Special emphasis is given to topics and problems of interest to the civil engineer. The work-energy method is used extensively in the solution of problems in kinetics.

Texts: Seely and Ensign, "Analytical Mechanics"; Joffe, "Problems in Mechanics".

Mech 23 KINEMATICS AND KINETICS. Prerequisites: Mech 20, Math 21.

The general aim and content of this course is the same as that of Mech 21. Special emphasis is given to topics and problems of interest to the mechanical engineer. The study of relative motion is extended to include Coriolis' Law.

Texts: Seely and Ensign, "Analytical Mechanics"; Joffe, "Problems in Mechanics".

Mech 24 STATICS, KINEMATICS AND KINETICS. Prerequisites: Math 21, Phys 3.

It is the aim of this course to acquaint the student of engineering with the fundamental laws, principles, and methods of mechanics, and to develop in him the ability to apply them in the solution of a great variety of problems of practical importance to the engineer. The principal topics included in this course are:

STATICS—Composition and resolution of forces and couples; equilibrium; analysis of simple frameworks; the laws of friction with general applications, and special reference to journal, belt and pivot friction.

KINEMATICS—linear and angular displacement, velocity, and acceleration; rectilinear and curvilinear motion; motion curves; relative motion; motion of rigid bodies.

KINETICS—Newton's laws applied to the motion of a particle; D'Alembert's principle; motion of the mass-center; translation, rotation and plane motion of a rigid body; work, power, energy, impulse, and momentum; principles of work and energy, principles of impulse and momentum, and their application to special types of motion of rigid bodies.

Texts: Seely and Ensign, “Analytical Mechanics”: Joffe, “Problems in Mechanics”.

DEPARTMENT OF PHYSICS

PROFESSOR F. N. ENTWISLE

ASSOC. PROF. E. SMITH

MR. W. HAZELL, JR.

MR. P. NIELSEN

MR. A. ZENTGRAF

The Department of Physics is in charge of the course in Physics given to Freshmen and Sophomores, and of the course in Strength of Materials given to Juniors.

It is the objective of the course in Physics to provide a knowledge of the fundamentals of the subject and to teach these fundamentals as prerequisite to later work in professional subjects rather than as basic principles in a discreet scientific subject. To this end the engineering aspects of the subject are stressed more than would be the case in General College Physics.

The schedule of instruction includes a rather small amount of formal lecturing with a large amount of informal recitation and problem work together with one afternoon each week spent in Laboratory. Effort is made to unify the instruction in the class room and in the laboratory. The work in the latter, which is largely quantitative, is designed to present fresh problems for the students' solution as far as possible rather than to require routine rechecking of known constants.

The course in Strength of Materials is designed to present the fundamental causes of the strain in material under stress. Effort is made to present a course which may be of common benefit to each of the four professional departments in the College. Instruction is carried out by means of lectures and recitations and one-half day per week spent in the Laboratory. The laboratory is designed to demonstrate the theory presented in the class room and thus furnish visual evidence of the accuracy of theoretical assumptions.

The Physics Laboratory

The Physics Department is supplied with two laboratories, adequately equipped with standard and special apparatus for quantitative measurements in elementary mechanics, heat, sound, light, and electricity. Its furnishings include sensitive physical balances, acceleration apparatus, coincidence and compound pendulums, Young's Modulus and centrifugal force apparatus, force tables, ballistic pendulums, microscopes, radiation equipment, and specially designed equipment for obtaining centers of gravity and moments of inertia of various specimens.

Equipment is at hand for performing standard experiments in heat. Kundt's tubes, electric tuning forks and resonance tubes are provided for experiments in sound. Measurements in light employ diffraction gratings, prisms, lenses, and mirrors. A first class optical bench with Lummer-Brodhun head, 30" sphere photometer, illuminometer, and Weston Photronic cells are used for illumination measurements. An equipment of meters, resistance units, Wheatstone bridges, potentiometer sets and traction permeameter is provided for elementary electrical measurements.

Strength of Materials Laboratory

The Strength of Materials Laboratory has been designed for the purpose of student instruction. With this in mind, the size of the apparatus has been kept within moderate limits so that the student may perform the test. The laboratory is housed in two adjoining rooms. The first room is equipped with:

- 1—100,000 lb Olsen Std. Tension Compression Test Machine.
- 1—50,000 lb Machine of the same sort.
- 1—50,000 lb hand operated Riehle Tension Compression Machine fitted with extra size screw for column work.
- 1—5000 lb Riehle Machine.
- 2—Punch Presses for shear studies.
- 2—Sets of Apparatus for testing Eccentric Riveted Joints.
- 1—Gyration Pendulum.
- 1—Polarized Light Stress Analyzer.
- 1—Torsion Demonstrator.
- 1—Slender Column Tester, also

Brinell Hardness Meter

Shore Scleroscope

Portable Brinell Tester

Extensometers, Strain Gauges, Shear Testers, for the above machines.
Torsion Meter, Planimeters, etc.

In the second room the cement and concrete testing appliances have been concentrated, including:

1—Riehle Briquette Testing Machine

Concrete Cylinder Moulds

Concrete Beam Moulds

Vicat Needles

Briquette Moulds, etc.

Moist Storage Cabinet

Sand and Gravel Bins

1—Power Concrete Mixer

SUBJECTS OF INSTRUCTION in the DEPARTMENT OF PHYSICS

Phys 1 INTRODUCTORY PROBLEMS IN PHYSICS.

An introductory course to familiarize the student with the best methods and procedure in performing calculations in Physics. Practice is given in the use of the slide rule, logarithms, mathematical and physical tables, construction of graphs and curves, co-ordinate and tabular ruled paper. Emphasis is placed upon the arrangement of work, efficiency of calculations and methods of attack. The question of precision is introduced through simple measurements and calculations and is emphasized throughout the work of the year. A set of problems has been compiled which aims to present the elementary principles of physics as basic to all engineering problems.

The work of the second term continues this approach, with special emphasis upon the proper preparation of reports in Physics. All of this work is done under conditions which approximate the environment of the engineering computing office.

Phys 2 and Phys 3. GENERAL PHYSICS.

The objective of the courses in General Physics is a knowledge of the fundamental laws of physical science, visualized as the foundation for later professional work. To this end, the courses are administered from the Engineering rather than the Scientific viewpoint.

PHYS 2.

ELEMENTARY MECHANICS—Linear and curvilinear motion; simple force system; energy and power; static forces in fluids; simple harmonic motion.

The laboratory work which accompanies this course is entirely quantitative and is designed to aid, by physical demonstration the development of the concepts originated in the classroom. To this end, the laboratory experimental work follows as closely as possible after the classroom exercises so that the essential unity of the two may be impressed upon the student's mind. An effort is made to develop the student's capacity for

sustained careful observation and deduction, and to initiate good practice in the matter of recording and reporting upon scientific and engineering data. Great stress is placed upon the precision of the results obtained in the laboratory.

Texts: "Physics", edited by Duff; Entwisle, "Experiments in Mechanics".

PHYS 3

HEAT, ELECTRICITY, SOUND AND LIGHT.

HEAT. Heat as a form of energy; calorimetry; expansion principles; heat transfer; meteorology.

ELECTRICITY. Fundamental principles of electric charge and electric current; development of essential mechanical nature of electrical and magnetic measurements.

SOUND. Wave motion; propagation; principles of sound quality; acoustics of rooms.

LIGHT. Illumination; photometry; principles of reflection; elementary geometrical optics; formation of spectra; interference; polarized light.

Laboratory work is given in the second semester of the course and covers a wide range of physical measurement, with particular attention given to the accuracy possible with the apparatus used.

Texts: "Physics", edited by Duff; Entwisle, "Experiments in Heat, Sound, Light and Electricity"; Entwisle, "Elements of Sound and Light".

Phys 30 STRENGTH OF MATERIALS. Prerequisites: Math 21, Mech 20.

The object of a study of Strength of Material is:

First, to determine the relations between the external forces acting on a body and the internal forces or stress and between external forces and the deformations or strains, so that the stresses may be determined from known loads or from measured strains or the strains determined from known loads.

Second, to obtain a knowledge of those properties of engineering materials necessary to an understanding of these relations.

Among the topics covered are stress-strain curves, properties of engineering materials, thin-walled cylinders, riveted joints, combined stresses and

strains, torsion, statically determinate and statically indeterminate beams, shear diagrams, moment diagrams, elastic curves, flexure formula, Euler column formula, Gordon-Rankine formula, straight line column formula, repeated loads, fatigue of metals, impact and energy loads, stresses in flat plates, and reinforced concrete beams.

An introduction to the use of a handbook is accomplished by instruction in the A. I. S. C. handbook and by the assignment of special problems for solution in class under supervision.

In the laboratory, tests are performed to verify the theoretical considerations studied in the classroom work. These include a study of testing machines, tension test of metal, test of riveted joint, compression tests, Brinell hardness, wood tests, strength of cement and mortar, concrete in bond and tension, construction and test of a reinforced concrete beam, slender column tests, torsion in shafts, and stress analysis by means of polarized light.

Texts: Frost, "Laboratory Manual"; Seely, "Resistance of Materials"; A. I. S. C. Handbook.

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